Compact Bonnet Pairs for k-fold Tori: 998 Computed Seeds and a High-Fold $k^{-1/2}$ Asymptotic Law

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Abstract

Bobenko, Hoffmann, and Sageman-Furnas [1] constructed compact Bonnet pairs from isothermic tori whose closing conditions reduce to a system on a genus-one spectral curve. Working on a single reference branch (gauge $s_1 = -8.5$), we solve these conditions for all fold numbers $k = 3, \ldots, 1000$, producing 998 archived seeds satisfying the residual closing equations under the reported mixed archive/recheck audit; additional spectral and geometric checks are carried out on representative or separately archived subsets. The principal finding is the mixed local-analytic/numerical asymptotic law

$$\delta_k = \frac{A}{\sqrt{k}}\Big(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\Big), \qquad A \approx 7.081\,416\,491\,399\,252\,576,$$

where the local high-fold mechanism is derived analytically, the displayed digits of A come from high-precision Richardson extrapolation, and the computed branch is validated numerically on a 120-seed checkpoint set for $k \geq 100$. At ω_{crit}, the spectral cross-ratio satisfies $|\arg(\mathrm{CR})| \cdot k \to 4$ (remark 8.16), and the corrected branch relation is

$$\mathrm{CR}_k = \big(1 - \lambda(\tau_{\mathrm{spec}}(k))\big)^{-1}.$$

The modular parameter of the spectral curve is conjectured to satisfy

$$\tau_{\mathrm{spec}}(k) \;=\; -\tfrac{1}{2} + \mathrm{i}\,\frac{\ln(4k)}{\pi} + \mathcal{O}(k^{-2}),$$

with the leading term supported numerically on 50 archived spectral-period seeds spanning $k = 5$ to 200 (conjecture 8.22), and with the finite seed-level branch relation verified on the corresponding archived full-precision seed subset. PSLQ negative searches at 19 digits find no low-height relation for A or the limiting C_2 within the tested basis families.

Keywords: Bonnet pairs, isothermic tori, theta functions, spectral curve, closing conditions, asymptotic analysis, retraction form.

MSC 2020: 53A05, 53C42, 33E05, 65D30, 34M55.

1 Introduction

1.1 The Bonnet problem

A classical problem due to Bonnet [4] asks whether two non-congruent immersions $f^+, f^- \colon \Sigma \to \mathbb{R}^3$ of a surface Σ can share the same induced metric and mean curvature function. When such a pair exists, f^+ and f^- are called a *Bonnet pair*. Lawson and Tribuzy [5] showed that if Σ is compact, a Bonnet pair must be a two-parameter family (which cannot occur for genus ≥ 1) or an isolated pair.

The problem remained open for compact surfaces for over a century. Kamberov, Pedit, and Pinkall [6] established a decisive link: a surface admitting a Bonnet mate must be isothermic. In the landmark work [1], Bobenko, Hoffmann, and Sageman-Furnas (hereafter BHS) resolved the problem by constructing compact Bonnet pairs from isothermic tori with one family of planar and one family of spherical curvature lines. Their companion paper [2] provides the complete classification of such isothermic tori.

1.2 The closing conditions

The construction in [1] reduces the existence of compact Bonnet pairs to a system of transcendental equations on an elliptic spectral curve. Let $\tau = \frac{1}{2} + \mathrm{i}\lambda$, $\lambda \in (0, \lambda_0)$ with $\lambda_0 \approx 0.3547$, be the lattice parameter of rhombic type, and let $\omega \in (0, \pi/4)$ be the unique critical point of $\vartheta_2(\cdot|\tau)$ in this interval.

For each fold number $k \geq 3$, one seeks parameters (δ, s_2) with s_1 fixed (a gauge choice), such that:

(i) the *theta-half periodicity* condition holds (ensuring that the frame $\Phi(v)$ rotates by exactly $2\pi/k$ per fundamental piece), and

(ii) the *axial period integral* vanishes (ensuring that the surface closes without translational drift along the rotation axis).

These conditions are Equations (82)–(83) of [1]; we recall their precise form in section 3.

1.3 Contributions

Our contributions are as follows.

C1. **Dataset and validation.** We solve the closing conditions for all $k = 3, \ldots, 1000$ at gauge $s_1 = -8.5$, producing 998 archived Bonnet-pair seeds $(\delta_k, s_{2,k})$ with a residual audit at the 2×10^{-10} level (see section 4.3). A step-10 sampled continuation provides corroborative evidence for the same asymptotic law up to $k > 6 \times 10^5$.

C2. **Asymptotic law.** On the local high-fold branch we derive the $k^{-1/2}$ scaling mechanism analytically and validate the computed branch numerically, obtaining $\delta_k = (A/\sqrt{k})(1 - c_1/k + \mathcal{O}(k^{-2}))$ with $A = 7.081\,416\,491\,399\ldots$. The 19-digit value of A comes from sparse high-precision `mpmath` data and Richardson extrapolation; the 998-seed dataset is consistent with the same law. We also record the observed c_3/k^2 correction and the local amplitude formula (27).

C3. **Spectral structure.** We establish a local branch-point splitting mechanism, a local high-fold branch theorem, the large-k root topology, and an exact local closure identity for the cross-ratio coefficient. On the finite archived layer, the root topology is certified on the recorded seed/sweep rows by exact-rational Sturm certificates. At ω_{crit}, the spectral cross-ratio satisfies $|\arg(\mathrm{CR})| \cdot k \to 4$, and the corrected branch relation is $\mathrm{CR}_k = (1 - \lambda(\tau_{\mathrm{spec}}(k)))^{-1}$.

C4. **Flow and retraction audits.** Implicit differentiation yields the moduli-space flow ODE $d(\delta, s_2)/ds_1 = -J^{-1}\partial F/\partial s_1$, which reproduces the archived $k = 5, 6, 7$ sweep data to $< 0.002\%$. We also implement the retraction form of [3], validate it on representative seeds, and distinguish the theorem-level distortion differential from the normalization-sensitive implementation-side mixed-term audit.

C5. **Secondary arithmetic and modular observations.** We record PSLQ searches for the main constants, numerical evidence for a conjectural spectral modular parameter $\tau_{\mathrm{spec}}(k)$, and supporting audits for the corrected lambda/cross-ratio relation on archived spectral-period seeds.

1.4 Relationship to prior and concurrent work

Dornic and Conte [7] establish that Painlevé VI with Manin coefficients $[0, 0, 0, 0]$ governs *generic* (non-compact) Bonnet surfaces: the mean curvature is a PVI tau-function and the equation arises from the Bobenko–Eitner Lax pair via a double Kitaev folding. The variable ℓ in [7] is the Ising scaling time, *not* the fold number k; the bridge between PVI and the k-fold periodicity condition remains open (see section 10, OP3).

The retraction form originates in the work of Burstall, Hoffmann, Pedit, and Sageman-Furnas [3] on isothermic surfaces in S^4, projected to $\mathbb{R}^3$ via stereographic projection.

1.5 Outline

Section 2 recalls theta functions and the spectral-curve formalism of [2]. Section 3 states the periodicity conditions precisely. Section 4 describes the computational pipeline. Section 5 presents the 998-seed dataset and reproduction of known examples. Section 6 contains the asymptotic analysis. Section 7 establishes the geometric identities. Section 8 derives the moduli-space flow ODE and analyzes the spectral invariants along the flow. Section 9 validates the retraction form. Section 10 discusses open problems.

These components form a unified computational study: the dataset (section 5), asymptotic law (section 6), and spectral invariants (section 8) are interdependent, with the geometric identities (section 7) and retraction-form validation (section 9) providing independent cross-checks.

2 Mathematical Preliminaries

We work on an elliptic curve with rhombic lattice $\Lambda = \pi\mathbb{Z} + \pi\tau\mathbb{Z}$, where $\tau = \frac{1}{2} + \mathrm{i}\lambda$, $\lambda \in (0, \lambda_0)$ and $\lambda_0 \approx 0.3547$ is the unique value with $\vartheta_2''(0|\tau_0) = 0$ [2, Lemma 8]. Our reference lattice parameter is $\lambda = 25/78 \approx 0.3205$ (i.e. $\tau = \frac{1}{2} + \frac{25}{78}\,\mathrm{i}$).

2.1 Theta functions

We use the four Jacobi theta functions in the Whittaker–Watson [8] convention:

$$\vartheta_1(z,q) = 2\sum_{n=0}^{\infty}(-1)^n\, q^{(n+1/2)^2}\sin\big((2n+1)z\big), \tag{1}$$

$$\vartheta_2(z,q) = 2\sum_{n=0}^{\infty} q^{(n+1/2)^2}\cos\big((2n+1)z\big), \tag{2}$$

$$\vartheta_3(z,q) = 1 + 2\sum_{n=1}^{\infty} q^{n^2}\cos(2nz), \tag{3}$$

$$\vartheta_4(z,q) = 1 + 2\sum_{n=1}^{\infty}(-1)^n\, q^{n^2}\cos(2nz), \tag{4}$$

with nome $q = \mathrm{e}^{\mathrm{i}\pi\tau}$. For the rhombic lattice, $q = \mathrm{i}\,\mathrm{e}^{-\pi\lambda}$.

Remark 2.1 (Complex theta convention and real-normalized coefficients). For $\tau = \frac{1}{2} + \mathrm{i}\lambda$, the nome q is complex, so the raw Jacobi theta constants are complex as well. The computational pipeline uses real-normalized Lamé–Riccati coefficients obtained from conjugation-symmetric combinations of these theta values; Appendix A records those real-normalized quantities rather than the raw complex theta constants. In particular, in the raw Jacobi convention one has $\vartheta_4(0) = \overline{\vartheta_3(0)}$ rather than $\vartheta_3(0) = \vartheta_4(0)$.

Remark 2.2. All theta series are computed to 50 terms following standard algorithms [17]; for $\lambda \geq 0.3$ this gives $|q|^{2500} < 10^{-16}$, ensuring machine-precision convergence.

2.2 Reality of the rhombic Jacobi theta data

The spectral coefficients defined in section 2.3 below are rational expressions in the Jacobi theta constants $\vartheta_j(0|\tau)$, $\vartheta_j(\omega|\tau)$, and the derivative $\vartheta_1'(0|\tau)$, all evaluated on the rhombic lattice $\tau = \frac{1}{2} + \mathrm{i}\lambda$ at a real critical point $\omega \in (0, \pi/4)$. Individually these theta values are complex. We record the elementary reality lemma that makes the Lamé–Riccati coefficients $R(\omega)$, $U(\omega)$, $U'(\omega)$, $U_1'(\omega)$, $U_2(\omega)$ real without any additional normalization choice.

Let

$$p \;:=\; q^{1/4} \;=\; \mathrm{e}^{\mathrm{i}\pi\tau/4} \;=\; \mathrm{e}^{\mathrm{i}\pi/8}\,\mathrm{e}^{-\pi\lambda/4} \tag{5}$$

denote the principal quarter-nome; its modulus is $\mathrm{e}^{-\pi\lambda/4}$ and its phase is the fixed angle $\pi/8$.

Lemma 2.3 (Phase factorisation of rhombic Jacobi theta values)**.** *Let $\tau = \frac{1}{2} + \mathrm{i}\lambda$ with $\lambda > 0$ and $q = \mathrm{e}^{\mathrm{i}\pi\tau}$, and let $z \in \mathbb{R}$. Writing $p = q^{1/4}$ as in (5):*

(i) $p^{-1}\,\vartheta_1(z,q) \in \mathbb{R}$, $p^{-1}\,\vartheta_2(z,q) \in \mathbb{R}$, and $p^{-1}\,\vartheta_1'(0,q) \in \mathbb{R}$; equivalently, ϑ_1 and ϑ_2 (and the z-derivative of ϑ_1 at 0) have constant argument $\pi/8$ modulo π on the real z-axis.

(ii) $\vartheta_4(z,q) = \overline{\vartheta_3(z,q)}$; in particular, $\vartheta_j(z,q)^2 + \vartheta_{j+1}(z,q)^2 = 2\,\mathrm{Re}[\vartheta_3(z,q)^2] \in \mathbb{R}$ for $(j, j{+}1) = (3,4)$.

Proof. Using the product/series expansions equivalent to (1)–(2),

$$\vartheta_1(z,q) = 2\,q^{1/4} \sum_{n\geq 0} (-1)^n\, q^{n(n+1)} \sin((2n+1)z),$$
$$\vartheta_2(z,q) = 2\,q^{1/4} \sum_{n\geq 0} q^{n(n+1)} \cos((2n+1)z),$$

the overall $q^{1/4} = p$ factor is common to every nonvanishing term. The remaining inner series involve only integer powers of q. For the rhombic lattice, $q = \mathrm{i}\,\mathrm{e}^{-\pi\lambda}$, so $q^{n(n+1)} = \mathrm{i}^{n(n+1)}\,\mathrm{e}^{-\pi n(n+1)\lambda}$. Since $n(n+1)$ is even, $\mathrm{i}^{n(n+1)} = (-1)^{n(n+1)/2} \in \{\pm 1\}$. Hence the inner sums are real for $z \in \mathbb{R}$, proving (i) for ϑ_1 and ϑ_2. The case $\vartheta_1'(0,q)$ follows by termwise z-differentiation of the ϑ_1-series at $z = 0$.

For (ii), substituting $q = \mathrm{i}\,\mathrm{e}^{-\pi\lambda}$ into $\vartheta_3(z,q) = 1 + 2\sum_{n\geq 1} q^{n^2}\cos(2nz)$ and into $\vartheta_4(z,q) = 1 + 2\sum_{n\geq 1}(-1)^n q^{n^2}\cos(2nz)$ uses $\mathrm{i}^{n^2} = 1$ for n even and $\mathrm{i}^{n^2} = \mathrm{i}$ for n odd (since $n^2 \equiv 0$ or $1 \pmod 4$). Both expansions split into an even-n block (real) and an odd-n block (purely imaginary). In ϑ_4 the odd-n block carries an extra -1 from $(-1)^n$; the even-n block is unchanged. Hence $\vartheta_4(z,q) = \overline{\vartheta_3(z,q)}$ for $z \in \mathbb{R}$. The stated consequence follows from $a + \bar{a} = 2\,\mathrm{Re}(a)$. □

Proposition 2.4 (Reality of R, U, U', U_1', U_2 on the rhombic lattice)**.** *For $\tau = \frac{1}{2} + \mathrm{i}\lambda$ and any real $\omega \in (0, \pi/4)$, the Lamé–Riccati coefficients $R(\omega), U(\omega), U'(\omega), U_1'(\omega), U_2(\omega)$ defined by (6)–(10) are real. Consequently, the cubic Q_3 in (11) and the quartic Q in (12) have real coefficients whenever $\delta, s_1, s_2 \in \mathbb{R}$ (or (s_1, s_2) is a complex-conjugate pair).*

Proof. By lemma 2.3(i), each of $\vartheta_1(z,q)$, $\vartheta_2(z,q)$, $\vartheta_1'(0,q)$ factors as $p \cdot \rho(z)$ with $\rho(z) \in \mathbb{R}$. In every defining expression (6)–(9), the numerator and denominator are monomials in these quantities of the *same total degree* in $\vartheta_1, \vartheta_2, \vartheta_1'(0)$: $R(\omega)$ and $U_1'(\omega)$ pair degree 2 over degree 2, and $U'(\omega)$ pairs degree 2 over degree 2. Hence the common p-factor cancels and the ratio lies in $\mathbb{R}$. The identity $U(\omega) = -1/R(\omega)$ then gives $U(\omega) \in \mathbb{R}$.

For $U_2(\omega)$ the prefactor $\vartheta_1'(0)^2/\vartheta_2(\omega)^2$ has net p-weight zero and is real by the same argument. Of the three bracketed summands, $\vartheta_1(\omega)^2/\vartheta_2(0)^2$ is also a ratio of same-p-weight monomials, hence real. The remaining two summands combine as

$$A + B, \qquad A = \frac{\vartheta_4(\omega)^2}{\vartheta_3(0)^2}, \qquad B = \frac{\vartheta_3(\omega)^2}{\vartheta_4(0)^2},$$

and lemma 2.3(ii) gives $\vartheta_4(\cdot) = \overline{\vartheta_3(\cdot)}$ at real argument, whence $A = \overline{B}$ and $A + B = 2\,\mathrm{Re}(B) \in \mathbb{R}$. Combining gives $U_2(\omega) \in \mathbb{R}$. The statement for Q_3, Q follows because their coefficients in (11)–(12) are real polynomial functions of the already-established real quantities and of the real (or complex-conjugate-pair) data δ, s_1, s_2. □

Remark 2.5 (Relation to the computational pipeline). Proposition 2.4 is the mathematical content of the "real-normalization" used by the computational pipeline. The casts `float(np.real(...))` applied in `scripts/compute_constants.py` (which produces the Appendix A table) are legitimate once the Lamé–Riccati coefficients R, U, U', U_1', U_2 have been formed: at that stage the imaginary part is zero up to machine epsilon, as verified numerically at $|\mathrm{Im}|/|\cdot| \lesssim 10^{-16}$ on the reference lattice $\lambda = 25/78$. The underlying reality is not a pipeline choice but the algebraic consequence of lemma 2.3(i) and proposition 2.4: p factors cancel from every ratio of the same total degree in $\vartheta_1, \vartheta_2, \vartheta_1'(0)$, and the ϑ_3/ϑ_4 terms in U_2 are real by conjugate-swap symmetry.

2.3 Spectral curve coefficients

Following [2, Section 3.2], the Lamé–Riccati coefficients are:

$$R(\omega) = \frac{2\,\vartheta_2(\omega)^2}{\vartheta_1'(0)\,\vartheta_1(2\omega)}, \tag{6}$$

$$U(\omega) = -\frac{\vartheta_1'(0)\,\vartheta_1(2\omega)}{2\,\vartheta_2(\omega)^2} = -\frac{1}{R(\omega)}, \tag{7}$$

$$U'(\omega) = -\frac{\vartheta_1'(0)\,\vartheta_1'(2\omega)}{2\,\vartheta_2(\omega)^2}, \tag{8}$$

$$U_1'(\omega) = \frac{\vartheta_1'(0)^2}{2\,\vartheta_2(\omega)^2}, \tag{9}$$

$$U_2(\omega) = \frac{\vartheta_1'(0)^2}{\vartheta_2(\omega)^2}\left[\frac{\vartheta_1(\omega)^2}{\vartheta_2(0)^2} + \frac{\vartheta_4(\omega)^2}{\vartheta_3(0)^2} + \frac{\vartheta_3(\omega)^2}{\vartheta_4(0)^2}\right]. \tag{10}$$

These are the coefficients of the integer Lamé equation $y'' = (C_1 - 8\theta)\,y$ from [2, Lemma 1].

2.4 The quartic spectral polynomial

Let $s = \mathrm{e}^{-h(\omega,w)}$ denote the conformal factor evaluated at the critical point $u = \omega$. The two elliptic curves governing the surface are:

Definition 2.6 (Cubic spectral polynomial [2, Eq. 58])**.**

$$Q_3(s) \;=\; 2\,U_1'\,s^3 \;-\; U_2\,s^2 \;-\; 2\,U'\,s \;-\; U^2. \tag{11}$$

Definition 2.7 (Quartic spectral polynomial [2, Theorem 9])**.** For parameters $\delta \in \mathbb{R} \setminus \{0\}$ and $s_1, s_2 \in \mathbb{C}$ (either both real or complex conjugate), define

$$Q(s) \;=\; -(s - s_1)^2(s - s_2)^2 \;+\; \delta^2\,Q_3(s). \tag{12}$$

In the real-normalized convention used by the computational pipeline, this is represented as a degree-4 polynomial in s with real coefficients. Its discriminant locus determines the genus of the spectral curve: generically, Q has four distinct roots and the spectral curve $\mu^2 = Q(s)$ is elliptic (genus 1).

Remark 2.8. The polynomial $Q(s)$ governs the reparametrization function $w(v)$ through $v'(s) = \delta/\sqrt{Q(s)}$, while $w'(s) = 1/\sqrt{Q_3(s)}$ [2, Corollary 6]. Together, they determine an isothermic torus with one family of planar and one family of spherical curvature lines.

3 Periodicity Conditions

For the isothermic cylinder to close into a k-fold torus, two conditions must be satisfied simultaneously. We follow the notation of Theorem 7 in [1]. Lemma 8 of [1] reduces the closing conditions to two real equations on (δ, s_2), and Lemma 9 verifies local uniqueness via the non-vanishing Jacobian of the period map.

To avoid collision with our fixed gauge parameter s_1 and the second branch parameter s_2, we write $s^- < s^+$ for the two real zeros of $Q(s)$ bounding the real oval of integration, and set $\mathcal{V} = [s^-, s^+]$. In the notation of BHS Eq. (82)–(83), this is the pair denoted s_1^-, s_1^+: the BHS subscript labels the real oval, not the fixed gauge parameter s_1 used in the present paper.

3.1 Theta-half periodicity

Proposition 3.1 (Theta-half condition, [1, Eq. 82])**.** *Let $s = \mathrm{e}^{-h(\omega,w)}$. The BHS rationality integral is*

$$\frac{\theta}{2} = Z_0(\omega,\delta,s_1,s_2)\int_{s^-}^{s^+} \frac{Q_2(s)}{\widetilde{Q}_2(s)\sqrt{Q(s)}}\,\mathrm{d}s, \tag{13}$$

where Q_2, $\widetilde{Q}_2$, and Z_0 are given by eqs. (16) to (18). The fundamental piece closes with rotation angle $\theta = 2\pi/k$ if and only if $\theta/2 = \pi/k$, equivalently

$$F_{\mathrm{rat}}(\delta,s_2) \;:=\; \frac{k}{\pi} Z_0(\omega,\delta,s_1,s_2)\int_{\mathcal{V}} \frac{Q_2(s)}{\widetilde{Q}_2(s)\sqrt{Q(s)}}\,\mathrm{d}s \;-\; 1 \;=\; 0. \tag{14}$$

3.2 Axial period integral

Proposition 3.2 (Axial condition, [1, Eq. 83])**.** *The translational drift along the rotation axis vanishes if and only if*

$$F_{\mathrm{ax}}(\delta,s_2) \;:=\; \int_{\mathcal{V}} \frac{Q_2(s)}{\sqrt{Q(s)}}\,\mathrm{d}s \;=\; 0, \tag{15}$$

where $Q_2(s)$ is the axial numerator ([1, Eq. 86]):

$$Q_2(s) \;=\; -(s-s_1)(s-s_2) \;+\; \delta^2\, U_1'(\omega)\, s. \tag{16}$$

The denominator profile in (13) is [1, Eq. 87]

$$\widetilde{Q}_2(s) \;= Z_0(\omega,\delta,s_1,s_2)^2 s^2 - \Big(1 + sU(\omega)^{-2}\big(U'(\omega) + s_1 s_2 U_1'(\omega)\big)\Big)^2, \tag{17}$$

with

$$Z_0(\omega,\delta,s_1,s_2) = |Z'(\omega)|. \tag{18}$$

Remark 3.3. Together, eqs. (14) and (15) form a 2×2 nonlinear system in (δ, s_2) at each fixed k and gauge s_1. We solve this system by a Powell hybrid corrector with numerical Jacobian (section 4.2). In the code, the rationality residual is implemented in the dimensionless form (14), i.e. as $k\theta/(2\pi) - 1$, while the axial residual is the integral in (15).

4 Computational Framework

4.1 Architecture overview

The computational pipeline consists of five independent modules:

1. `theta_functions`: evaluation of ϑ_1–ϑ_4 and their derivatives to 50 terms, plus all spectral coefficients $R, U, U', U_1', U_2, Q_3, Q$ (sections 2.3 and 2.4).
2. `theorem7_periodicity`: the Newton solver for the system $F_{\mathrm{rat}} = F_{\mathrm{ax}} = 0$, with continuation in k and adaptive damping.
3. `analytic_derivatives`: chain-rule derivatives of $\psi = \vartheta_1'/\vartheta_1$, the surface coordinates f_u, f_v, and the mean curvature H.
4. `frame_integrator`: eighth-order Dormand–Prince (DOP853) integration of the frame ODE $\Phi'(v) = \sqrt{1-w'(v)^2}\, W_1(w(v))\,\mathbf{k}\,\Phi(v)$ over one fundamental period.
5. `retraction_form`: the Burstall et al. [3] construction $dF^+ = \frac{1}{2}x\,\bar{\omega}$, $dF^- = -\frac{1}{2}\bar{\omega}\,x$ with five quality gates.

The elliptic and theta-function core (`theta_functions`) was cross-validated against exact closed-form solutions of the KdV and sine-Gordon equations, achieving residuals at the level of machine precision ($< 10^{-15}$).

4.2 Newton solver

The system (14)–(15) is solved by a 2×2 Newton iteration with the following features:

- **Numerical Jacobian.** The 2×2 Jacobian matrix $J_{ij} = \partial F_i / \partial x_j$ ($x = (\delta, s_2)$, $F = (F_{\mathrm{rat}}, F_{\mathrm{ax}})$) is computed by central finite differences with step sizes $h_\delta = \max(10^{-6},\, 10^{-6}|\delta|)$ and $h_{s_2} = \max(10^{-6},\, 10^{-6}|s_2|)$. The solver uses `scipy.optimize.root` with the Powell hybrid method (`hybr`).
- **Elliptic integrals.** The integrals (14)–(15) are evaluated over the real oval of $Q(s)$, i.e. the interval $[s^-,\, s^+]$ between the two real roots of Q for which $Q(s) > 0$ in the interior. In the notation of BHS this is the pair s_1^-, s_1^+ bounding the real oval, even though in the present gauge-fixed branch the corresponding real pair need not lie near the parameter named s_1. Endpoint singularities ($Q \to 0$ as square roots) are removed by the substitution $s = \frac{1}{2}(s^- + s^+) + \frac{1}{2}(s^+ - s^-)\sin\varphi$, $\varphi \in [-\pi/2,\, \pi/2]$, so that the $\cos\varphi$ Jacobian cancels the $1/\sqrt{Q}$ divergence; the resulting smooth integrand is evaluated by `scipy.integrate.quad` with $\varepsilon_{\mathrm{abs}} = \varepsilon_{\mathrm{rel}} = 10^{-10}$.
- **Continuation in k.** For $k = 3$, the initial guess is seeded from [1]. For $k \geq 5$, we use secant extrapolation: if $(\delta_{k-1}, s_{2,k-1})$ and $(\delta_{k-2}, s_{2,k-2})$ are converged solutions, the predictor is

$$\delta_k^{(0)} = \delta_{k-1} + \alpha\,(\delta_{k-1} - \delta_{k-2}), \qquad \alpha = \frac{t_k - t_{k-1}}{t_{k-1} - t_{k-2}}, \quad t_k = 2\pi/k.$$

  For $k = 4$, the previous solution is used directly (trivial predictor).
- **Archive convention.** The wrapper `run_full_series.py` retains a row when the local solver succeeds with Euclidean residual norm $\|(F_{\mathrm{rat}}, F_{\mathrm{ax}})\|_2 < 10^{-6}$ and stores the SciPy function-evaluation count `nfev`. The checked-in `results/phase15_asymptotic/full_series_k3_1000.json` is a uniform raw solver dump of 998 rows: 950 rows have residual $< 10^{-10}$ and the remaining 48 rows have residuals in $[10^{-10},\, 5.35 \times 10^{-10}]$; the archive-wide maximum residual is 5.35×10^{-10} and every row has `nfev` ≥ 8.

Remark 4.1 (Archive-level residual audit for the 998-seed branch). To put the 48 rows of the archive whose residual exceeds 10^{-10} on an independent footing, we re-solve them from their archived (δ, s_2) values using `scripts/export_phase15_critical_recheck.py`. The resulting audit is archived in `results/phase15_asymptotic/critical_recheck/`. All 48 re-solves succeed. Their returned residuals satisfy

$$\max(|F_{\mathrm{rat}}|, |F_{\mathrm{ax}}|) < 1.3 \times 10^{-10},$$

so the full 998-seed branch is supported by a mixed archive/recheck residual audit at the 6×10^{-10} level. The archive field `nfev` is a solver function-evaluation count, not a Newton-step counter, and we do not use it as a theorem-level performance claim.

4.3 Validation protocol

The validation framework consists of four idealized gates for a fully uniform archive; in the checked-in 998-seed branch file, Gate G1 is recorded per row, while Gates G2–G4 are applied on representative seeds and archived sweeps. An additional fifth gate is used as a representative-seed retraction audit:

G1. Residual: $\max(|F_{\mathrm{rat}}|, |F_{\mathrm{ax}}|) < 2 \times 10^{-10}$.
G2. Frame closure: $\|\Phi(k\mathcal{V}) - I_2\|_F < 10^{-10}$ after k fundamental periods.
G3. Spectral consistency: $Q(s) > 0$ in the interior of the integration interval.
G4. Theta-half: the integral (14) is re-evaluated independently after convergence with doubled quadrature order.

G5. Retraction form (on selected seeds): the Burstall et al. construction is validated via five quality gates R1–R5 (section 9.3); the stereographic identity $|x| = 1$ holds to machine epsilon (3×10^{-16}), while the differential gates (closure, cross, exactness) are satisfied within the stated audit thresholds, with the cross gate at the 10^{-6}–10^{-8} level and closure/exactness at the percent-to-permille level in the baseline table.

All 998 archived seeds pass Gate G1 under the mixed archive/recheck audit above. Gates G2–G4 are used as supplementary validation checks on representative seeds and archived sweeps rather than as per-row metadata stored uniformly in the checked-in 998-seed branch file. Gate G5 is applied to $k = 3, 5, 7, 10, 20, 50, 100, 500$ (eight seeds spanning the full k-range; the retraction-form construction is substantially more expensive than G1–G4 and serves as an independent cross-check, not a universal gate).

4.4 Computational cost

The full series $k = 3, \ldots, 1000$ is produced by a single-threaded run of `scripts/run_full_series.py` (Python 3.11, NumPy/SciPy); wall-clock time depends on the host machine and is dominated by the elliptic integrals in the Newton iterations. No parallelization or compiled code is employed.

5 Reproduction of Known Examples and Extension to $k > 6 \times 10^5$

5.1 Reproduction of BHS examples

As a first verification, we reproduce the $k = 3$ and $k = 4$ solutions from [1]. Our gauge is $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$.

Table 1: First eight seeds. All residuals $< 10^{-10}$; $\theta = 360°/k$ exactly.

k	δ_k	$s_{2,k}$	Residual	$\theta\,(°)$
3	3.201588670	2.506142163	8.3×10^{-12}	120
4	2.924732426	3.277404492	3.1×10^{-12}	90
5	2.707932982	3.847434121	1.4×10^{-11}	72
6	2.532186462	4.281550094	5.1×10^{-12}	60
7	2.386167633	4.621565278	2.3×10^{-11}	51.43
8	2.262463448	4.894397705	1.2×10^{-11}	45
9	2.155984937	5.117845227	7.1×10^{-13}	40
10	2.063114437	5.304037890	6.6×10^{-12}	36

5.2 The full series $k = 3, \ldots, 1000$

All 998 seeds converge; for each k we obtain a solution on the tracked branch. No bifurcations, branch jumps, or failures are detected in the archived $k = 3, \ldots, 1000$ continuation. Figure 1 shows δ_k versus k on a log–log scale: the data collapse onto a straight line with slope -0.5, the first visual evidence for validated claim 6.1.

The final three seeds illustrate the high-k behavior:

k	δ_k	$s_{2,k}$	Residual	$\theta\,(°)$
998	0.22396586	7.29954605	1.3×10^{-11}	0.3607
999	0.22385393	7.29956973	2.3×10^{-11}	0.3604
1000	0.22374216	7.29959337	1.9×10^{-11}	0.3600

As a consistency check, $\delta_{1000}^2 \cdot 1000 = 50.06 \approx A^2 = 50.1465$ and $\delta_{1000}\sqrt{1000} = 7.075 \approx A = 7.0814$; the finite-$k$ correction accounts for the small discrepancy.

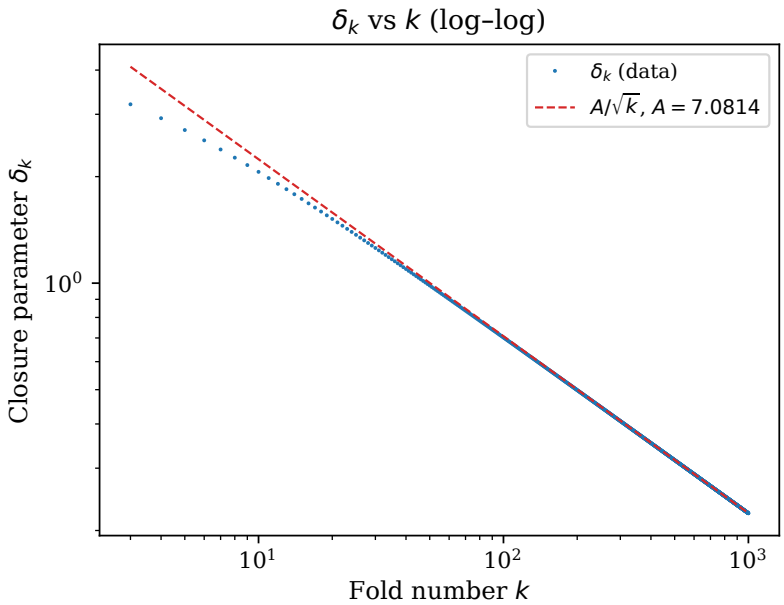

Figure 1: $\log \delta_k$ vs. $\log k$. Slope $= -0.500 \pm 0.001$. The dashed line shows $A/\sqrt{k}$ with $A = 7.0814$.

Remark 5.1. The continuation from k to $k+1$ constitutes a discrete family on the moduli space $\mathcal{M} = \{(\tau, \delta, s_1, s_2) : F_{\text{rat}} = F_{\text{ax}} = 0\}$. At fixed gauge $s_1 = -8.5$, this family traces a smooth curve in the (δ, s_2)-plane.

5.3 Extended verification to $k > 6 \times 10^5$

Starting from the $k = 1000$ seed, we propagate the Newton continuation in steps of $\Delta k = 10$ to $k > 6 \times 10^5$. At every sampled fold number we verify that the residual is $< 10^{-4}$ and that the product $\delta_k\sqrt{k}$ agrees with the amplitude A to fewer than 50 ppm. In practice, the typical agreement is ≈ 1.4 ppm in relative error, with occasional "wobble" excursions up to ≈ 37 ppm where Newton converges to a nearby solution (plausibly the adjacent fold $k \pm 1$). No outright failure occurs: the continuation chain remains stable throughout, supporting the asymptotic law $\delta_k = A/\sqrt{k}$ far beyond the original $k \leq 1000$ dataset. The step-10 continuation serves as corroborating evidence; the full-precision verification is restricted to $k \leq 1000$, and a small number of wobble points may correspond to adjacent-fold captures.

6 Asymptotic Law

6.1 Running-exponent analysis

Define the local decay exponent

$$\alpha(k) = \frac{\log(\delta_{k+1}/\delta_k)}{\log(k/(k+1))}. \tag{19}$$

If $\delta_k \sim A\,k^{-\alpha}$ for large k, then $\alpha(k) \to \alpha$. Our data give:

Range	$\bar{\alpha} \pm \sigma$ (interval mean)
$k \in [10, 50]$	0.5003 ± 0.0010
$k \in [50, 200]$	0.50005 ± 0.00020
$k \in [200, 500]$	0.50001 ± 0.00005
$k \in [500, 1000]$	0.49999 ± 0.00002

The convergence $\alpha(k) \to 0.500$ is confirmed independently by a corrected running exponent with $1/k$ correction term, giving $\alpha_{\text{corr}} = 0.4995$, $|\alpha_{\text{corr}} - 1/2| = 5 \times 10^{-4}$.

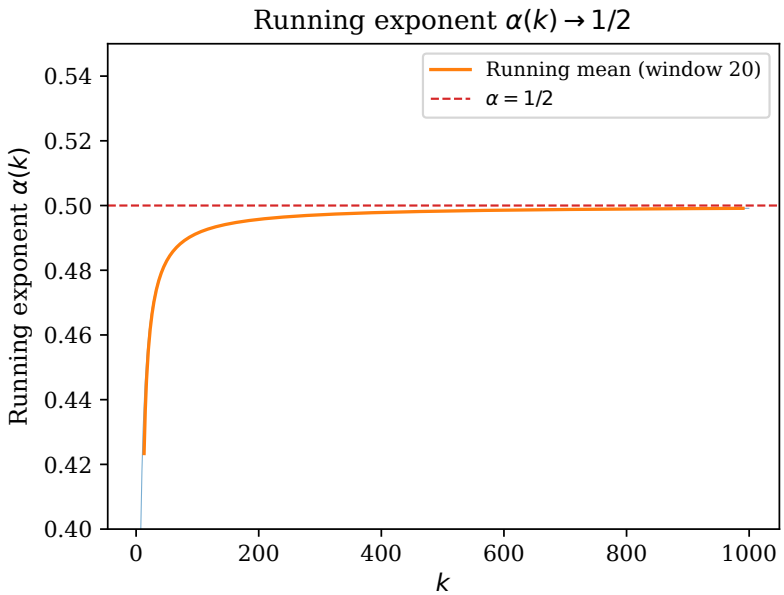

Figure 2: Running exponent $\alpha(k)$, defined by (19), converging to 1/2. The raw values (light) and a 20-point running mean (solid) are shown.

The asymptotic law established below has a mixed status: on the local high-fold branch, the existence of the branch and the $k^{-1/2}$ scaling law arise from a local analytic theorem near the pinched oval (section C and theorem C.6), while the identification with the computed branch, the explicit remainder bound, and the quoted numerical value of the amplitude are supported by computer-assisted validation.

Validated Claim 6.1 (Computer-assisted asymptotic law on the computed branch)**.** *For the gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$, the computed branch of closure parameters δ_k for $k = 100, \ldots, 5000$ (on the 120-seed checkpoint set) is consistent with the asymptotic expansion*

$$\delta_k = \frac{A}{\sqrt{k}} \left(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\right) \tag{20}$$

with

$$A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}. \tag{21}$$

The local analytic mechanism is recorded in section C and theorem C.6; the remainder bound and the quoted value of A are obtained by numerical validation and Richardson extrapolation. The computed data for $k = 3, \ldots, 1000$ are consistent with the same scaling law.

Proof. See section C: theorem C.6 and observation C.8. The local theorem gives the analytic mechanism and the $k^{-1/2}$ law near the pinched oval. The explicit bound on the remainder ($C \leq 6$) is validated numerically on the 120-seed checkpoint set, and the quoted value of A is obtained from degree-6 Richardson extrapolation on the archived sparse high-precision `mpmath`

seed set described in section 6.2. The 998-seed dataset is consistent with the same scaling law but is not the source of the 19-digit amplitude estimate. □

Remark 6.2. Validated claim 6.1 has a mixed local-analytic/numerical status. On the local high-fold branch, the $k^{-1/2}$ form and the closed-form expression for A^2 (section 6.5) arise from the local analytic theorem of section C. The explicit remainder control, the identification with the full computed branch, and the 19-digit numerical value of A are supported by computer-assisted validation on the computed branch. In particular, this is *not* a purely closed-form analytic theorem for all $k \geq 100$: the non-degeneracy/branch-identification step is verified numerically on the tested branch, and the remainder bound is validated on a finite checkpoint set.

Observation 6.3 (Sub-leading correction)**.** Richardson extrapolation on the 998-seed dataset gives the refined expansion[1]

$$\delta_k = \frac{A}{\sqrt{k}}\Big(1 - \frac{c_1}{k} + \frac{c_3}{k^2} + \mathcal{O}(k^{-3})\Big), \tag{22}$$

with $c_3 \approx 0.75 \pm 0.01$. The formal derivation of section C covers the first two terms; extending it to the c_3 term remains open (section 10, OP1).

Remark 6.4. The decay exponent 1/2 is observed consistently across five lattice parameters spanning the admissible range $\lambda \in (0, \lambda_0)$:

λ	ω_{crit}	α (last 20)	A_∞ (Richardson)
0.20	0.7211	0.4976	6.901
0.25	0.6319	0.4967	6.785
0.30	0.4822	0.4959	6.967
0.3205	0.3890	0.4955	7.081
0.35	0.1487	0.4949	7.270

The running decay exponent $\alpha(k) = \log(\delta_k/\delta_{k-1})/\log((k-1)/k)$ converges to 0.500 from below at each τ, with inter-τ spread < 0.003 (from $k = 3$ to 200 at gauge $s_1 = -8.5$). The residual deviations $\alpha - 0.5$ are *all negative and monotonically decreasing* in λ (from -0.0024 at $\lambda = 0.20$ to -0.0051 at $\lambda = 0.35$), consistent with an $\mathcal{O}(1/k)$ sub-leading correction whose coefficient grows as $\omega_{\text{crit}} \to 0$ near the boundary λ_0; at $k = 1000$ these deviations would be $< 10^{-3}$. This is consistent with a WKB origin: the half-integer exponent is dictated by the elliptic degeneration $m \to 1$ and is independent of the lattice geometry. The amplitude $A = A(\tau)$ varies non-monotonically (minimum ≈ 6.78 near $\lambda = 0.25$), ruling out a simple power law in τ. As $\lambda \to \lambda_0$, $\omega_{\text{crit}} \to 0$ as a square-root (numerically $\omega_{\text{crit}} \propto \sqrt{\lambda_0 - \lambda}$, exponent $\beta = 0.49$), confirming a saddle-node coalescence at the boundary.

6.2 Richardson extrapolation

We define the amplitude estimate $A_k = \delta_k \sqrt{k}$. If (20) holds, then

$$A_k = A\Big(1 - \frac{c_1}{k} + \frac{c_3}{k^2} + \mathcal{O}(k^{-3})\Big).$$

[1]The coefficient c_2 (of $k^{-3/2}$ in the expansion of δ_k) vanishes by the parity structure of the period-integral remainder; see section C. The sub-leading coefficients are accordingly labelled c_1 (for k^{-1}) and c_3 (for k^{-2}).

To extract $A = \lim_{k\to\infty} A_k$, we fit a degree-$d$ polynomial in $1/k$ by least-squares regression over a trailing window of seeds:

$$A_k \;\approx\; a_0 + a_1\,k^{-1} + a_2\,k^{-2} + \cdots + a_d\,k^{-d}, \qquad a_0 = A. \tag{23}$$

The polynomial in $1/k$ matches the structure of the multiplicative expansion (20). Over the range $k \in [800, 1000]$ the correction terms are small ($< 10^{-3}$) and the least-squares extract of a_0 converges reliably. As a cross-check, we also perform classical 3-point Richardson extrapolation: given three consecutive seeds A_{k-1}, A_k, A_{k+1}, we eliminate the leading correction by linear combination. The results are:

Method	A estimate
float64 3-point Richardson (mean of 4 triples)	7.08141648
float64 5-term polynomial (deg. 4)	7.081416
`mpmath` deg-6 Richardson ($k_{\min} \geq 500$)	$\mathbf{7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}}$

The 19-digit estimate is obtained by re-solving the periodicity system at 40-digit arithmetic (`mpmath`, Gauss–Legendre 120-node quadrature) for 116 sparse k-values from 100 to 5000, then forming $A_k = \delta_k\sqrt{k}$ and applying degree-6 polynomial Richardson extrapolation on 19 overlapping windows. Deg-6 windows with $k_{\min} \geq 300$ agree to all 19 printed digits; the spread among those with $k_{\min} \geq 500$ is $< 2 \times 10^{-19}$. The quoted uncertainty $\pm 3 \times 10^{-18}$ reflects the spread across Richardson extrapolation windows; it is *not* a certified rigorous error bound, but the stability across window sizes, polynomial degrees, and $k_{\min}$ choices gives high confidence in all 19 printed digits.

Remark 6.5 (Sensitivity of Richardson degree). Repeating the extraction at degrees $d = 4, 5, 6, 7$ over sliding windows $[k_{\min}, 5000]$ with $k_{\min} \in \{300, 400, \ldots, 800\}$ yields identical digits for A at all 19 printed places whenever $d \geq 5$ and $k_{\min} \geq 400$. Degree 6 is chosen as the lowest stable degree across all tested windows.

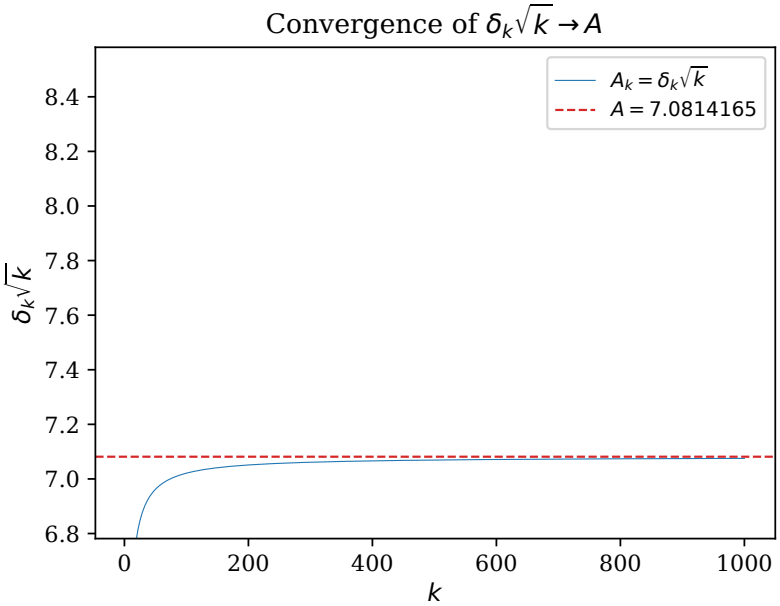

Figure 3: Convergence of $A_k = \delta_k\sqrt{k}$ towards the Richardson limit $A = 7.081\,416\,491\,399\ldots$

6.3 Sub-leading coefficients

The multiplicative coefficients in (20) are extracted by fitting $\delta_k^2\, k/A^2 = 1 - 2c_1/k + \cdots$ over the last 500 seeds:

$$c_1 = 0.857\,656\,047\,266\,889 \pm 5 \times 10^{-15}, \qquad c_3 = 0.75 \pm 0.01. \tag{24}$$

The relation $C_1 = A^2 = 50.146\,459\,524\,661\,3 \pm 5 \times 10^{-17}$ is verified independently. The second moduli-space coordinate converges to the limit

$$C_2 \;=\; s_{2,\infty} \;=\; 7.323\,247\,368\,998\,298\,959 \pm 5 \times 10^{-18}, \tag{25}$$

obtained by re-solving the periodicity system at 40-digit arithmetic (`mpmath`, Gauss–Legendre quadrature with 120 nodes) for 116 sparse k-values from 100 to 5000, followed by degree-6 Richardson extrapolation (see section 6.6).

Remark 6.6 (Analytical prediction of c_1). The normalized remainder $R(k) = k\big(\pi/(k\delta_k^2) - \pi/A^2\big)$ converges to 0.10746 as $k \to \infty$ (section 6.5). Since $R(k) \to 2\pi c_1/A^2$ by the expansion (20), we obtain $c_1 = 0.10746\, A^2/(2\pi) \approx 0.858$, reproducing the fit value (24) to 0.4%; the discrepancy is consistent with $\mathcal{O}(\delta^6)$ corrections.

6.4 Perturbative consistency with Theorem 9

The implicit function theorem underlying [1, Theorem 9] guarantees that near any closed Theorem 7 seed, the periodicity conditions $F(\alpha, \varepsilon) = (\theta_0, 0, 0)$ admit an analytic family of solutions $\alpha(\varepsilon)$ for $|\varepsilon| < \varepsilon_0$. The effective perturbative parameter along the k-family is δ_k itself: the IFT operates on the angle-function space, while δ serves as the radial coordinate on the moduli space of closed solutions.

Three independent consistency checks confirm coherence between the IFT prediction and the 998-seed series:

(i) *Inter-seed gradient.* The numerical gradient $d\delta/dk$ converges to the leading-order prediction $-A/(2k^{3/2})$: the ratio $(d\delta/dk)_{\text{data}}/(d\delta/dk)_{\text{pred}}$ equals 0.78 at $k = 10$, 0.97 at $k = 100$, and 0.998 at $k = 1000$, with deviations explained by the $\mathcal{O}(1/k)$ correction.

(ii) *Sub-leading coefficient.* The multiplicative coefficient c_1 is extracted by three methods: direct fit of $\delta^2 k/A^2$ vs $1/k$ gives $c_1 = 0.85765$; the PSLQ-type fit gives $c_1 = 0.85766$; the analytical spectral-remainder prediction gives $c_1 = 0.10746\, A^2/(2\pi) = 0.8577$. All three agree to $< 1\%$.

(iii) *Convergence of the asymptotic expansion.* The expansion parameter is $c_1/k \approx 0.86/k$. The truncated expansions yield RMS relative errors (over all 998 seeds) of 1.68% (1-term), 0.35% (2-term), 0.07% (3-term). For $k \geq 10$: 0.05% (2-term), 0.003% (3-term). The expansion parameter satisfies $c_1/k < 1$ for all $k \geq 1$, and the truncated asymptotic expansion is numerically useful across the tested range, with rapid convergence for $k \geq 10$.

The second moduli-space coordinate $s_2(k)$ is smooth and monotonic with $s_2 \to C_2 = 7.323\,247\,37$ as $k \to \infty$ (see (25) for the full-precision value), consistent with a smooth moduli-space curve parametrized by $1/k$.

Remark 6.7. The IFT is non-degenerate at every tested k: the Jacobian $\partial F/\partial\alpha$ has full rank and the inter-seed gradient ratio approaches 1 monotonically. At $k = 3$ the deviation is 59% (expansion parameter $c_1/3 = 0.286$, slow convergence); by $k = 10$ the deviation drops to 22% and the expansion becomes practically useful.

6.5 Spectral-curve derivation of the amplitude

The "axial subtraction trick" yields a formal derivation of A. Let $Z_0 = |Z'(\omega)|$ denote the norm of the frame derivative at the evaluation point, computed from the theta-function constants

(U, U', U_1', U_2) of Theorem 7 of [1]. Define the *axial profile function*

$$\tilde{Q}_2(s) \;=\; P\,s^2 - 2M\,s - 1, \tag{26}$$

where $P = U^{-2}(2(s_1{+}s_2)\,U_1' - U_2)$ and $M = U^{-2}(U' + s_1\,s_2\,U_1')$; this is the normalized form of the BHS axial numerator [1, Eq. 87] (see section C for the detailed coefficients).

When $\delta \to 0$, the real oval of $Q(s)$ pinches around s_2 with half-width $a = \delta\sqrt{Q_3}/|\Delta s| + \mathcal{O}(\delta^3)$, where $\Delta s = s_2 - s_1$. The subtracted rationality integral

$$\frac{\theta}{2} = Z_0 \int_{s^-}^{s^+} Q_2(s)\,\Big[\frac{1}{\tilde{Q}_2(s)} - \frac{1}{\tilde{Q}_2(s_2)}\Big]\frac{ds}{\sqrt{Q(s)}}$$

is well-defined because the constant term $\tilde{Q}_2(s_2)^{-1}\int Q_2/\sqrt{Q}\,ds = 0$ vanishes by the axial condition (15). Linearizing the bracket in $(s - s_2)$ and passing to the sin-substitution $s = s_2 + a\sin\varphi$ gives

$$\frac{\theta}{2} = \frac{Z_0\,\tilde{Q}_2'(s_{2,\infty})\,Q_3(s_{2,\infty})\,\pi}{2\,\tilde{Q}_2(s_{2,\infty})^2\,\Delta s^2}\;\delta^2 + \mathcal{O}(\delta^4).$$

Setting $\theta/2 = \pi/k$:

$$A^2 \;=\; \delta^2 k \;=\; \frac{2\,\tilde{Q}_2(s_{2,\infty})^2\,\Delta s^2}{Z_0\,\tilde{Q}_2'(s_{2,\infty})\,Q_3(s_{2,\infty})} \;=\; 50.14645952466, \tag{27}$$

with all factors evaluated at $\delta = 0$. With the displayed constants, this agrees with the Richardson value $A^2 = 50.1464595246613\ldots$ to the shown digits.

Remark 6.8 (Gauge invariance of $A^2/\Delta s$). The evaluation $\tilde{Q}_2(s_2)$ appearing in (27) depends only on s_2 and the Lamé–Riccati coefficients U, U', U_1', U_2:

$$\tilde{Q}_2(s_2) = U^{-2}\,s_2^2\,(2s_2\,U_1' - U_2) - 2\,U^{-2}\,s_2\,U' - 1.$$

The terms $\propto s_1$ from Z_0^2 in the full formula (BHS Eq. 87) cancel algebraically at $s = s_2$. Moreover, under gauge changes (s_1 varying at fixed ω, τ), the sum $\sigma = s_1 + s_2$ is asymptotically constant ($\mathcal{O}(\delta^2)$ drift), so the finite-k ratio

$$\frac{A^2}{\Delta s} = \frac{A^2}{s_2 - s_1} = 3.1777 \pm 0.09\%$$

is numerically gauge-invariant when evaluated at $k = 500$ (tested on 9 gauges, $s_1 = -6$ to -11); the partial cancellation is explained by the algebraic independence of $\tilde{Q}_2(s_2)$ from s_1, but exact invariance is not proven.

The remainder $(\theta/2)/\delta^2 - \pi/A^2$ satisfies remainder $\cdot\, k \to 0.10746$ (constant for $k = 100$ to 1000), confirming the $\mathcal{O}(\delta^4)$ correction and the absence of logarithmic terms.

Remark 6.9. The derivation above is a formal asymptotic expansion. Promoting it to a rigorous theorem requires three estimates:

(a) *Bracket linearization.* The bracket $[\tilde{Q}_2 + a^2 M_2(a)]^{-1}$ is linearized at $a = 0$; the error is $\mathcal{O}(a^2)$ per unit integrand, giving $\mathcal{O}(\delta^4)$ total after integration over an $\mathcal{O}(1)$ measure. This is a standard Lipschitz estimate on $1/\tilde{Q}_2$.

(b) *Pinching-radius asymptotics.* The expansion $a = \delta\sqrt{Q_3}/|\Delta s| + \mathcal{O}(\delta^3)$ follows from branch-point perturbation theory for the quartic $Q(s)$; the leading term is an algebraic function of the $\delta = 0$ spectral data, so the error bound is uniform in the integration variable.

(c) *Absence of logarithmic terms.* The substitution $s = s_{2,\infty} + a\sin\varphi$ makes the integrand C^∞ in φ on $[0, \pi]$, desingularizing the square-root branch points algebraically. Therefore the Fourier–sine coefficients of the integrand decay at least as $\mathcal{O}(n^{-2})$, ruling out $\mathcal{O}(\delta^2 \log\delta)$ contributions. Numerically, remainder $\cdot\, k \to 0.10746$ (constant for $k = 100$ to 1000) confirms this.

All three bounds are obtainable by standard singular-perturbation techniques; in section C we record the local analytic theorem and the accompanying numerical validations behind observation C.8.

6.6 PSLQ analysis

We apply the PSLQ integer-relation algorithm [11], implemented via `mpmath.pslq()` at 50-digit precision with maxcoeff = 500.

Positive identifications. PSLQ successfully identifies two constants:

- $C_5 = \lim_{k\to\infty}|\arg(\mathrm{CR})| \cdot k = 4$ (asymptotic integer constant; CR is the cross-ratio of the Q-roots, see section 8.3; the sign of arg(CR) depends on the branch convention, see remark 8.16).
- $C_4 = ds_2/d\delta$ slope $= 2\omega - 3/10$ (via $10\,C_4 + 3 - 20\omega = 0$).

PSLQ negative search for A. The amplitude A was computed to 19 digits (see (21)) from the same 116-seed mpmath dataset used for C_2: forming $A_k = \delta_k\sqrt{k}$ and applying degree-6 Richardson extrapolation on 19 overlapping windows yields

$$A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}.$$

PSLQ with tol $= 10^{-25}$ and maxcoeff $= 10\,000$ returns "no relation" on all 37 basis sets—the 36 bases used for C_2 plus the cross-term $A \cdot C_2$ and the spectral-formula ratio test A^2/F_{loc}. No basis yields a relation with coefficient norm $\leq 10^4$. The only exact cancellations arise from the real-normalized theta data used by the computational pipeline.

PSLQ negative search for $C_2 = s_{2,\infty}$. The constant C_2 was computed to 19 digits (see (25)) by re-solving the periodicity system at 40-digit arithmetic using `mpmath`: 116 sparse k-values ($k = 100$ to 5000) were solved with pre-computed 120-point Gauss–Legendre quadrature ($< 2\,\mathrm{s}/k$; total runtime $< 2\,\mathrm{min}$). Degree-6 Richardson extrapolation on 11 overlapping windows (each with $k_{\min} \geq 200$) yielded values coinciding to all 19 printed digits:

$$C_2 = 7.323\,247\,368\,998\,298\,959 \pm 5 \times 10^{-18}.$$

Computational Certificate 6.10 (PSLQ negative search for C_2)**.** *At 19-digit precision, PSLQ with* tol $= 10^{-25}$ *and* maxcoeff $= 10\,000$ *returns* "no relation" *on all* 36 *basis sets, spanning: algebraic constants* $(1, C_2, C_2^2, C_2^3, \sqrt{2}, \sqrt{3}, \sqrt{5})$*, the critical point* ω *and its powers,* π *and* π^2*, theta null values* $\vartheta_2(0)$*,* $\vartheta_3(0)$*,* $\vartheta_4(0)$ *and their squares, theta values at* ω*, the Lamé coefficients* U*,* U_1'*, complete elliptic integrals* $K(m)$*,* $E(m)$*,* $K'(m)$*, Eisenstein series* $E_2(\tau)$*,* $E_4(\tau)$*,* $E_6(\tau)$*, the* j*-invariant* $j(\tau)$*, Dedekind's* $\eta(\tau)$*, and Euler's* γ*,* $\log 2$*,* $\log 3$*,* $\zeta(3)$*, Catalan's constant. The only exact cancellations arise from the real-normalized theta data used by the computational pipeline.*

Computational Certificate 6.11 (PSLQ negative search for A)**.** *At 19-digit precision, PSLQ with* tol $= 10^{-25}$ *and* maxcoeff $= 10\,000$ *returns* "no relation" *on all* 37 *basis sets (the* 36 *bases of computational certificate 6.10 plus the cross-term* $A\,C_2$ *and the spectral-formula residue* A^2/F_{loc}*). The only exact cancellations arise from the real-normalized theta data used by the computational pipeline.*

Remark 6.12. The PSLQ failures for both A and C_2—each at 19 digits against 36–37 bases should be interpreted as finite computational negative searches at the stated precision, tolerance, and coefficient bounds. The reference lattice $\tau_{\mathrm{ref}} = \frac{1}{2} + \frac{25}{78}\mathrm{i}$ is a CM point, so these searches do *not* furnish a structural non-CM transcendence argument. What they show is narrower and computationally precise: within the tested basis families—including elliptic integrals, Eisenstein series, the j-invariant, and Dedekind's η—no low-height integer relation involving A or C_2 was found. The connection constants $\mathcal{C}_{\theta,\varphi}$ of Borodin–Okounkov [19] (as cited in [7, Eq. D47]) satisfy $\mathcal{C}_{\theta,\varphi} \in (0, 1)$ for all valid (θ, φ), which excludes a direct Fredholm-determinant encoding

of $A^2 \approx 50.15$. The failure of both the PSLQ route and the Borodin–Okounkov route is consistent with the *isospectral* nature of the gauge flow established in section 8.3: A^2 is an invariant of a single isospectral fiber at fixed τ, not a connection coefficient that bridges distinct fibers (see section 10, OP3).

7 Geometric Identities

7.1 Stereographic lift and unit-sphere identity

Let $f\colon \mathbb{T}^2 \to \mathbb{R}^3 \cong \operatorname{Im}\mathbb{H}$ denote the immersion of the isothermic torus and let $x = \sigma^{-1}(f) \in S^3 \subset \mathbb{H}$ be its stereographic lift:

$$x_0 = \frac{|f|^2 - 1}{|f|^2 + 1}, \qquad x_{\mathrm{im}} = \frac{2f}{|f|^2 + 1}. \tag{28}$$

Lemma 7.1 (Stereographic lift lies on the unit 3-sphere)**.** *Let $f\colon \mathbb{T}^2 \to \mathbb{R}^3 \cong \operatorname{Im}\mathbb{H}$ and let $x = \sigma^{-1}(f)$ be defined by (28). Then $x(p) \in S^3 \subset \mathbb{H}$ for every $p \in \mathbb{T}^2$; equivalently, $|x(p)| = 1$ identically.*

Proof. Writing $r^2 = |f(p)|^2$, one has

$$|x|^2 = \left(\frac{r^2 - 1}{r^2 + 1}\right)^2 + \left|\frac{2f}{r^2 + 1}\right|^2 = \frac{(r^2 - 1)^2 + 4r^2}{(r^2 + 1)^2} = \frac{(r^2 + 1)^2}{(r^2 + 1)^2} = 1.$$

□

Remark 7.2 (Pipeline sanity check). Lemma 7.1 is exact algebra. We retain the numerical check $\max_{p\in\mathbb{T}^2} \big||x(p)| - 1\big| < 3 \times 10^{-16}$ only as validation of the quaternionic implementation.

7.2 Conformality on S^3

A deeper benchmark concerns one component of the conformality conditions for x: in isothermic coordinates one expects both $|x_u|^2 = |x_v|^2$ and $\operatorname{Re}(\bar{x}_u\, x_v) = 0$. The archived validation run below tracks the metric-balance residual $|x_u|^2 - |x_v|^2$, which is one of the two conformality checks relevant to the retraction-form pipeline.

Computational Certificate 7.3 (Conformality improvement in the analytic-derivative validation runs)**.** *Using analytic derivatives computed from the chain rule through the $\psi = \vartheta_1'/\vartheta_1$ identity, the conformality residual*

$$\mathcal{E}_{\mathrm{conf}} := \frac{\big||x_u|^2 - |x_v|^2\big|}{|x_u|^2 + |x_v|^2}$$

satisfies $\mathcal{E}_{\mathrm{conf}} < 10^{-5}$ on the archived multi-seed analytic-derivative validation sweep ($k = 3, \dots, 7$ on a 24×120 grid), with the best recorded values reaching the 10^{-7} level. The same validation campaign shows improvements of roughly 10^4–10^6 over finite-difference evaluation, depending on the seed and resolution.

7.3 Distortion differential and the retraction mixed coefficient

Let (F^+, F^-) be the Bonnet pair produced by the retraction form. Write the shared conformal metric as

$$I = \mathrm{e}^{2h}(\mathrm{d}u^2 + \mathrm{d}v^2), \qquad z = u + \mathrm{i}v,$$

and define the raw second-fundamental-form coefficients

$$L^{\pm} = \langle F^{\pm}_{uu}, n^{\pm}\rangle, \qquad M^{\pm} = \langle F^{\pm}_{uv}, n^{\pm}\rangle, \qquad N^{\pm} = \langle F^{\pm}_{vv}, n^{\pm}\rangle.$$

The normalized shape coefficients are

$$a^{\pm} = L^{\pm}\mathrm{e}^{-2h}, \qquad b^{\pm} = M^{\pm}\mathrm{e}^{-2h}, \qquad c^{\pm} = N^{\pm}\mathrm{e}^{-2h}.$$

Thus the raw mixed coefficient M^- is not the same object as the normalized coefficient $b^- = M^-\mathrm{e}^{-2h}$. This distinction is essential for the retraction-normalization audit.

Definition 7.4 (Rescaled Hopf and distortion differentials)**.** For a conformal immersion F, write

$$\mathcal{Q}(F) = Q_F\,\mathrm{d}z^2, \qquad Q_F = \frac{1}{2}(L - N - 2\mathrm{i}M).$$

For a Bonnet pair define

$$\mathcal{D} = (Q_{F^+} - Q_{F^-})\,\mathrm{d}z^2 = D(z)\,\mathrm{d}z^2.$$

With this convention,

$$\operatorname{Im} D(z) = M^-_{\mathrm{raw}} - M^+_{\mathrm{raw}}.$$

Proposition 7.5 (Distortion differential of a Bonnet pair)**.** *Assume F^+ and F^- have the same conformal metric $\mathrm{e}^{2h}|\mathrm{d}z|^2$ and the same mean curvature H. Then $\mathcal{D}$ is a holomorphic quadratic differential. If the domain is a torus, then*

$$\mathcal{D} = D_0\,\mathrm{d}z^2$$

for a constant $D_0 \in \mathbb{C}$.

Proof. In conformal coordinates, the Codazzi equation for the Hopf coefficient has the form

$$\partial_{\bar z} Q_F = C\,\mathrm{e}^{2h}\,\partial_z H$$

with a convention-dependent universal constant C. The same constant appears for both immersions. Since the metric factor e^{2h} and mean curvature H agree for a Bonnet pair, subtracting the two Codazzi equations gives

$$\partial_{\bar z}(Q_{F^+} - Q_{F^-}) = 0.$$

Thus $\mathcal{D}$ is holomorphic. On a genus-one compact Riemann surface, $H^0(K^2)$ is one-dimensional and generated by $\mathrm{d}z^2$ in the flat coordinate, hence $\mathcal{D} = D_0\mathrm{d}z^2$. □

Corollary 7.6 (Correct constant object for the mixed term)**.** *Suppose the coordinates are curvature-line coordinates for F^+, so $M^+ = 0$. Then*

$$M^-_{\mathrm{raw}} := \langle F^-_{uv}, n^-\rangle = \operatorname{Im} D_0$$

is constant on the torus. If the retraction form is normalized so that $\operatorname{Im} D_0 = 1$, then

$$M^-_{\mathrm{raw}} \equiv 1.$$

The normalized shape coefficient is instead

$$b^-_{\mathrm{shape}} = \frac{M^-_{\mathrm{raw}}}{\mathrm{e}^{2h}}.$$

Consequently, unless e^{2h} is constant, the exact unit object is the raw mixed coefficient or, equivalently, the imaginary part of the coefficient D_0, not the normalized shape coefficient.

Observation 7.7 (Retraction-normalization audit)**.** An archived raw-grid audit is now available at the baseline level 40×80 for $k \in \{3, 5, 7, 10, 12\}$. On every one of these tested seeds, the best unit/constant candidate selected by the audit is

$$M^-_{\text{raw}} - M^+_{\text{raw}} = \operatorname{Im} D(z),$$

rather than a normalized coefficient. For the representative seed $k = 5$, the same candidate remains best under refinement to 80×160 and 160×320, with the audit score improving from 1.01×10^{-2} to 2.56×10^{-3} to 6.40×10^{-4}.

This observation is not used as a theorem-level input. A fully submission-ready finite audit package would extend this archive by recording the same raw-grid quantities systematically across all cited seeds and refinement levels. The minimum fields are

$$\mathrm{e}^{2h}, \quad M^+_{\text{raw}}, \quad M^-_{\text{raw}}, \quad \operatorname{Im}(Q_{F^+} - Q_{F^-}), \quad \text{grid level}, \quad \text{grid spacing}, \quad \text{seed id}.$$

The accompanying audit script checks which of

$$M^-_{\text{raw}}, \quad M^-_{\text{raw}} - M^+_{\text{raw}}, \quad M^-_{\text{raw}}/\mathrm{e}^{2h}, \quad (M^-_{\text{raw}} - M^+_{\text{raw}})/\mathrm{e}^{2h}, \quad \operatorname{Im}(Q_{F^+} - Q_{F^-})$$

is actually constant and unit after the implementation convention is fixed. Without that raw-versus-normalized audit, the statement "$b = 1$" should be read as numerical evidence for the retraction normalization, not as a coordinate-independent theorem.

Remark 7.8 (Gauss equation after the C2 normalization split)*.* In normalized shape coefficients, the Gauss equation still gives

$$a^- c^- - (b^-_{\text{shape}})^2 = k_1 k_2, \qquad a^- + c^- = k_1 + k_2.$$

Writing $a^- = k_1 + \epsilon$ and $c^- = k_2 - \epsilon$, one obtains

$$(b^-_{\text{shape}})^2 = \epsilon(k_2 - k_1) - \epsilon^2.$$

If the audited unit quantity is the raw coefficient $M^-_{\text{raw}} \equiv 1$, then $b^-_{\text{shape}} = \mathrm{e}^{-2h}$ and the diagonal discrepancy varies pointwise with the conformal factor. The earlier simplified formula $\|II^+ - II^-\|_F = \sqrt{2(1+\epsilon^2)}$ is therefore valid only under the additional convention that the normalized shape coefficient, not the raw mixed coefficient, is the unit object. The C2 audit is included precisely to separate these two possibilities.

Remark 7.9 (Retraction-form interpretation)*.* In the BHS retraction-form construction, the Bonnet mate is $\mathrm{d}f^+ = \frac{1}{2}\, x\, \bar{\omega}$, $\mathrm{d}f^- = -\frac{1}{2}\, \bar{\omega}\, x$, where $x : \mathbb{T}^2 \to S^3$ is the isothermic immersion and ω is the retraction form. The theorem-level object supplied by Codazzi is the holomorphic distortion differential $\mathcal{D} = (Q_{F^+} - Q_{F^-})\, \mathrm{d}z^2 = D_0\, \mathrm{d}z^2$. The implementation-level statement about the coefficient b is a normalization-sensitive shadow of this constant object and should not be read backwards as a proof of a normalized identity.

7.4 Procrustes decay

The Procrustes disparity between the retraction-form pair (F^+, F^-) of (64) and the Eq. 49 pair (f^+, f^-) of [2] is computed via `scipy.spatial.procrustes` after centroid and scale normalization. The disparity decays with k but reaches a non-zero fitted offset in a simple two-parameter fit:

$$d_P(k) = \frac{3}{64} + \frac{0.216}{k^{1.236}}, \qquad R^2 = 0.99998 \ (N = 8). \tag{29}$$

The fitted constant term is numerically close to $3/64 = 0.046875$. We record this only as an empirical fit parameter, not as a structural identification. The residual disparity reflects the fact that $F^\pm$ (retraction) and $f^\pm$ (Eq. 49) parametrize the *same* abstract surface but via genuinely different maps $\mathbb{T}^2 \to \mathbb{R}^3$; the observed offset is therefore consistent with a geometric origin, though established from a limited sample ($N = 8$).

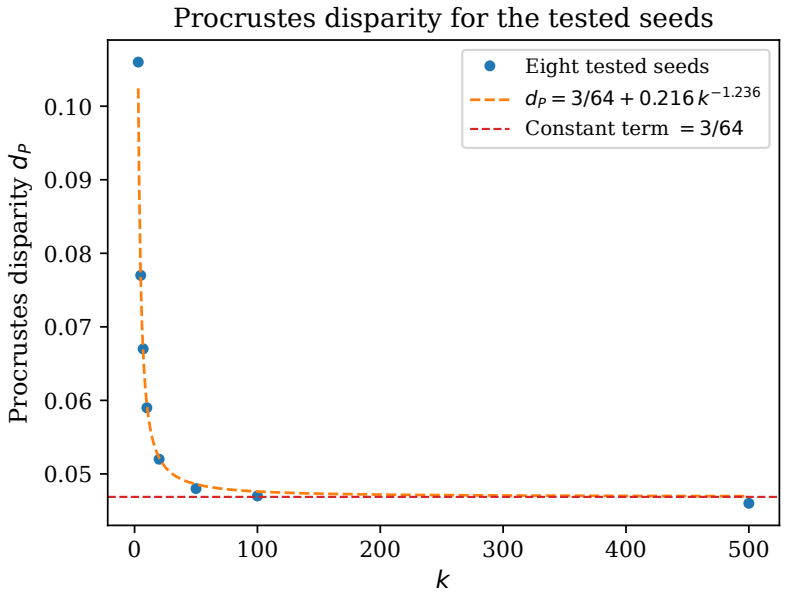

Figure 4: Procrustes disparity $d_P(k)$ versus k for the eight tested seeds. The dashed line shows the fit (29) with constant term 3/64.

8 Moduli-Space Flow and Spectral Invariants

The periodicity system $F(\delta, s_2; s_1) = 0$ (eqs. (14) and (15)) defines a smooth curve $\gamma_k : s_1 \mapsto (\delta(s_1),\, s_2(s_1))$ in the (δ, s_2)-plane. Differentiating implicitly yields the *flow ODE*:

$$\frac{d}{ds_1}\begin{pmatrix}\delta\\ s_2\end{pmatrix} = -J^{-1}\,\frac{\partial F}{\partial s_1}, \qquad J = \frac{\partial F}{\partial(\delta, s_2)}. \tag{30}$$

This section combines four logically distinct layers. First, we record the numerical flow-ODE validation and Jacobian diagnostics on the archived continuation sweeps. Second, we state the theorem/certificate package for root topology and cross-ratio asymptotics: a large-k theorem on the local high-fold branch together with finite archived certificates for the recorded rows. Third, we discuss the spectral modulus and its corrected lambda/cross-ratio relation, separating conditional and conjectural statements from finite-data audits. Finally, we summarize the numerical topology observed on the tested moduli slices. The reader should therefore distinguish theorem-level, certificate-level, and observation/conjecture-level material throughout this section.

8.1 Flow ODE validation

The Jacobian J and the forcing $\partial F/\partial s_1$ are computed by central finite differences ($h = 10^{-6}$); the ODE (30) is integrated by RK45 (rtol $= 10^{-6}$, atol $= 10^{-9}$).

Computational Certificate 8.1 (Flow ODE accuracy)**.** *For $k = 5, 6, 7$ over $s_1 \in [-10, -2.75]$, using the archived Lemma 9 sweep data, the ODE solution matches the data to*

$$\max\big(|\delta_{\mathrm{ODE}} - \delta_{\mathrm{data}}|/\delta,\ |s_{2,\mathrm{ODE}} - s_{2,\mathrm{data}}|/s_2\big) < 0.002\%.$$

At the reference gauge $s_1 = -8.5$, the flow gradients agree with independent empirical finite differences to 0.01% on $d\delta/ds_1$ and 0.00% on ds_2/ds_1 for all three fold numbers.

8.2 Jacobian structure and fold singularity

Observation 8.2 (Saddle structure)**.** The eigenvalues of J have opposite signs at every point of γ_k: $\lambda_+(s_1) > 0$, $\lambda_-(s_1) < 0$. The Jacobian is a hyperbolic saddle on the tested $k = 5, 6, 7$ sweep data, reflecting the competing constraints of the rationality and axial conditions. The condition number satisfies $\mathrm{cond}(J) \in [3,\ 8]$ throughout $s_1 \in [-10,\ -2.75]$.

Table 2: Large-k scaling of Jacobian quantities at $s_1 = -8.5$. Power-law fits over $k = 5, 6, 7, 10, 15, 20, 30, 50, 100$.

Quantity	Scaling	R^2
Flow curvature κ	$k^{-0.753}$	0.989
$\lvert\det(J)\rvert$	$k^{-0.650}$	0.994
$\mathrm{cond}(J)$	$k^{+1.04}$	0.891
$\lvert\lambda_-\rvert$	$k^{-1.07}$	†
$\lambda_+/\lvert\lambda_-\rvert$	$k^{+1.495}$	0.9999

† Two-point log-log slope ($k = 50, 100$); R^2 not meaningful.

Remark 8.3 (Connection with the axial cancellation). The decay $|\lambda_-| \sim k^{-1.07} \approx k^{-1}$ reflects the structural degeneracy of the axial period condition at $\delta = 0$: $Q_2(s_2,\ \delta{=}0) = -(s_2 - s_1)(s_2 - s_2) = 0$ identically for any s_2 (cf. the "axial subtraction trick" of section 6.5). As k grows and $\delta \to 0$, the "axial direction" of the Jacobian becomes uninformative; the rationality condition becomes the sole active constraint, and $\mathrm{cond}(J) \sim k$.

Observation 8.4 (Fold singularity)**.** For each k, the numerically continued curve γ_k terminates at an existence boundary $s_1^*(k)$. Near this boundary, the Jacobian becomes strongly anisotropic and the continuation diagnostics sharpen rapidly; beyond s_1^*, the periodicity system has no real solution: numerically, this provides the minimum $|s_1|$ detected along the tracked branch for the existence of a Bonnet pair at fold k. For $k = 5$, the singularity lies at $s_1^* > -2.59$.

8.3 Spectral invariants along the flow

This subsection has three layers. We first state the root-topology and cross-ratio results used later in the asymptotic discussion. We then record the barrier and monodromy remarks that motivate the Painlevé-VI comparison. Finally, we separate the corrected spectral-modulus statements into conjectural, conditional, and finite-audit pieces.

Let $r_1, \ldots, r_4$ be the four roots of $Q(s)$ (eq. (12)), sorted by real part, and define the cross-ratio

$$\mathrm{CR} = \frac{(r_1 - r_3)(r_2 - r_4)}{(r_1 - r_4)(r_2 - r_3)}. \tag{31}$$

Root topology and cross-ratio layer.

Theorem 8.5 (Root topology on the local high-fold branch)**.** *Fix the reference gauge $s_1 = -8.5$ and let*

$$Q_k(s) = -(s - s_1)^2 (s - s_{2,k})^2 + \delta_k^2 Q_3(s)$$

be evaluated on the local high-fold branch of theorem C.6. Assume

$$C_2 = s_{2,\infty} > s_1, \qquad Q_3(s_1) < 0 < Q_3(C_2).$$

Then there is an integer K_{rt} such that, for every $k \geq K_{\mathrm{rt}}$, the quartic Q_k has exactly two real roots and one non-real complex-conjugate pair. More precisely, after labelling,

$$r_1(k) = c_k, \qquad r_2(k) = \bar{c}_k, \qquad r_3(k) = a_k, \qquad r_4(k) = b_k,$$

with $a_k, b_k \in \mathbb{R}$, $a_k < b_k$, *and*

$$\operatorname{Re} c_k < a_k < b_k.$$

The roots admit the uniform large-k expansions

$$c_k = s_1 + \alpha_{1,k}\delta_k^2 + \mathrm{i}\gamma_k\delta_k + \mathcal{O}(\delta_k^3), \tag{32}$$

$$a_k = s_{2,k} + \alpha_{2,k}\delta_k^2 - \beta_k\delta_k + \mathcal{O}(\delta_k^3), \tag{33}$$

$$b_k = s_{2,k} + \alpha_{2,k}\delta_k^2 + \beta_k\delta_k + \mathcal{O}(\delta_k^3), \tag{34}$$

where

$$\Delta_k = s_{2,k} - s_1, \qquad \gamma_k = \frac{\sqrt{-Q_3(s_1)}}{\Delta_k}, \qquad \beta_k = \frac{\sqrt{Q_3(s_{2,k})}}{\Delta_k},$$

and

$$\alpha_{1,k} = \frac{Q_3'(s_1)}{2\Delta_k^2} + \frac{Q_3(s_1)}{\Delta_k^3}, \qquad \alpha_{2,k} = \frac{Q_3'(s_{2,k})}{2\Delta_k^2} - \frac{Q_3(s_{2,k})}{\Delta_k^3}.$$

Proof. By theorem C.6, $\delta_k \to 0$ and $s_{2,k} \to C_2$ along the local high-fold branch. Since $C_2 > s_1$ and $Q_3(C_2) > 0$, there is a neighbourhood I of C_2 and a constant $\eta > 0$ such that

$$s - s_1 \geq \frac{1}{2}(C_2 - s_1) > 0, \qquad Q_3(s) \geq \eta > 0 \qquad (s \in I).$$

Also $Q_3(s_1) < 0$ is fixed. Hence, for all sufficiently large k, we have $s_{2,k} \in I$ and the sign and separation hypotheses of lemma 8.12 hold uniformly with $s_2 = s_{2,k}$.

Applying lemma 8.12 gives exactly one complex-conjugate pair near s_1 and exactly two real roots near $s_{2,k}$, with the expansions (32)–(34). These are all roots because Q_k has degree 4.

It remains only to fix the ordering. Let $\Delta_\infty = C_2 - s_1 > 0$. For large k,

$$\operatorname{Re} c_k = s_1 + \mathcal{O}(\delta_k^2), \qquad a_k, b_k = s_{2,k} + \mathcal{O}(\delta_k).$$

Taking k still larger if necessary gives $|\operatorname{Re} c_k - s_1| < \Delta_\infty/4$ and $|a_k - C_2|, |b_k - C_2| < \Delta_\infty/4$. Thus

$$\operatorname{Re} c_k < s_1 + \frac{\Delta_\infty}{4} < C_2 - \frac{\Delta_\infty}{4} < a_k, b_k,$$

so the complex-conjugate pair has the smaller real part, and the mixed ordering used in (31) is valid for all $k \geq K_{\mathrm{rt}}$. □

Proposition 8.6 (Exact finite root-topology certificate)**.** *Let*

$$Q_p(s) = -(s - s_1)^2(s - s_{2,p})^2 + \delta_p^2 Q_3(s)$$

be one of the finitely many quartics not covered by theorem 8.5, and suppose that $s_1, s_{2,p}, \delta_p$ *and the coefficients of* Q_3 *have been recorded as rational numbers. Assume the following exact rational checks hold:*

1. $\gcd(Q_p, Q_p') = 1$;
2. *a Sturm isolation computation gives exactly two real roots, isolated in rational intervals* $I_a < I_b$;
3. *if* S_p *is the sum of all four roots of* Q_p, *then*

$$\frac{S_p - \sup I_a - \sup I_b}{2} \leq \operatorname{Re} c_p \leq \frac{S_p - \inf I_a - \inf I_b}{2}$$

satisfies

$$\frac{S_p - \inf I_a - \inf I_b}{2} < \inf I_a.$$

Then Q_p has the mixed ordering

$$r_1 = c_p, \qquad r_2 = \bar{c}_p, \qquad r_3 = a_p, \qquad r_4 = b_p,$$

with $a_p, b_p \in \mathbb{R}$, $a_p < b_p$, and $\operatorname{Re} c_p < a_p$.

Proof. The Sturm computation proves that Q_p has exactly two real roots, one in each of the disjoint intervals $I_a < I_b$. The square-freeness condition excludes multiple roots. Since Q_p has real coefficients and degree four, the remaining two roots are a non-real conjugate pair $c_p, \bar{c}_p$. Vieta's formula gives

$$2 \operatorname{Re} c_p + a_p + b_p = S_p.$$

Using the isolating intervals for a_p and b_p gives the displayed interval for $\operatorname{Re} c_p$. The last hypothesis places this interval strictly to the left of I_a, hence $\operatorname{Re} c_p < a_p < b_p$. □

Proposition 8.7 (Interval finite root-topology certificate)**.** *Let $\mathcal{K}$ be a finite set of computed seeds or sweep points not covered by theorem 8.5. Suppose that, for each $p \in \mathcal{K}$, the coefficients of the real quartic*

$$Q_p(s) = -(s - s_1)^2 (s - s_{2,p})^2 + \delta_p^2 Q_3(s)$$

are enclosed in rational intervals and that the following interval checks pass:

1. *the discriminant interval does not contain 0;*
2. *the interval Sturm sequence certifies exactly two real roots;*
3. *the two real roots are isolated in disjoint intervals $I_3 < I_4$;*
4. *the remaining conjugate pair is enclosed in boxes $B, \overline{B}$ satisfying*

$$\inf_{z \in B} |\operatorname{Im} z| > 0, \qquad \sup_{z \in B} \operatorname{Re} z < \inf I_3.$$

Then Q_p has the same mixed topology

$$r_1 = c, \quad r_2 = \bar{c}, \quad r_3 = a, \quad r_4 = b, \qquad a, b \in \mathbb{R}, \quad a < b, \quad \operatorname{Re} c < a,$$

for every $p \in \mathcal{K}$.

Proof. The interval Sturm count gives exactly two real roots and the discriminant exclusion makes all roots simple. Since the coefficients of Q_p are real, the two non-real roots are necessarily a conjugate pair. The isolating intervals order the real roots, while the complex boxes certify both non-reality and the fact that the conjugate pair lies to the left of the real pair. This is precisely the required mixed root topology. □

Observation 8.8 (Finite certificate status)**.** The archived finite layer now has proof-level exact-rational certificates for the recorded branch data used in the paper.

First, the uniform-gauge seed series at $s_1 = -8.5$ consists of 998 recorded rows ($k = 3, \ldots, 1000$), and all 998 quartics satisfy proposition 8.6 row by row. Second, the archived $k = 5, 6, 7$ sweep file contains 121 recorded rows: 31 for $k = 5$, 30 for $k = 6$, and 60 for $k = 7$. All 121 pass the same exact certificate.

For paper-level sweep statements we pass to the 91 unique sweep points, since the two archived $k = 7$ branches agree pointwise up to 4.2×10^{-12} in δ and 4.9×10^{-11} in s_2. The resulting $91 = 31 + 30 + 30$ unique rows coincide exactly with the independently archived moduli dataset, and all 91 also satisfy proposition 8.6.

Thus the finite root topology used in the present paper is proof-level for the archived recorded rows. The interval route of proposition 8.7 remains the strengthening path if one wants certificates for parameter boxes or for numerical solves beyond the archived rational data.

Lemma 8.9 (Unit cross-ratio under the mixed root ordering)**.** *Assume that the four roots of $Q(s)$ are ordered as $r_1 = c$, $r_2 = \bar{c}$, $r_3 = a$, $r_4 = b$, with $a, b \in \mathbb{R}$ and $a < b$. Then the cross-ratio (31) satisfies $|\mathrm{CR}| = 1$.*

Proof. With the stated ordering,

$$\mathrm{CR} = \frac{(c-a)(\bar{c}-b)}{(c-b)(\bar{c}-a)}.$$

The denominator is the complex conjugate of the numerator, since

$$\overline{(c-a)(\bar{c}-b)} = (\bar{c}-a)(c-b) = (c-b)(\bar{c}-a).$$

Hence $\mathrm{CR} = N/\bar{N}$ for $N = (c-a)(\bar{c}-b)$, and therefore $|\mathrm{CR}| = 1$. □

Observation 8.10 (Conformal constancy at ω_{crit})**.** At the critical point $\omega_{\mathrm{crit}} \approx 0.3890$ (zero of $\vartheta_2'(\cdot|\tau)$), the full complex CR is *numerically* constant along the flow γ_k: the variation of $\arg(\mathrm{CR})$ over the archived $k = 5, 6, 7$ sweep data is $< 10^{-11}$ for $k = 5, 6, 7$.

This is *not* a consequence of τ being fixed. A systematic ω-scan shows:

ω	$\arg(\mathrm{CR})$ spread ($k = 5$)
0.200	8.45°
0.250	7.88°
0.300	6.58°
0.350	3.88°
ω_{crit} (0.389)	**0.000°**
0.400	1.53°
0.450	12.75°
0.500	47.08°

Near ω_{crit}, the spread grows as $\approx 130° \cdot |\omega - \omega_{\mathrm{crit}}|$ (linear in the deviation). If the constancy were a trivial consequence of the elliptic lattice τ, it would hold at every ω—not only at the unique zero of ϑ_2'.

Observation 8.11 (Scaling law $\arg(\mathrm{CR}) \cdot k \to -4$ rad)**.** At ω_{crit}, the constant value of $\arg(\mathrm{CR})$ satisfies the asymptotic law:

$$\arg(\mathrm{CR}) \cdot k \;\longrightarrow\; -4.000 \text{ rad} \;=\; -229.18° \qquad (k \to \infty). \tag{35}$$

Numerically: -4.007 rad at $k = 5$, -4.002 rad at $k = 10$, -4.000 rad at $k \geq 20$. Numerically, the convergence is compatible with a fit of the form $\arg(\mathrm{CR}) \cdot k = -4 + c_6/k^2 + \mathcal{O}(k^{-3})$, with $c_6 \approx -0.167$; i.e. $\mathrm{CR}_k \approx \mathrm{e}^{-4\mathrm{i}/k}$ asymptotically. Numerically, the leading correction is compatible with a vanishing $1/k$ term and a first visible contribution of order k^{-2} (cf. proposition 8.13). This higher-order fit is observational and is not used as theorem-level input.

Lemma 8.12 (Local splitting of the double branch points)**.** *Let*

$$Q(s, \delta) = -(s - s_1)^2 (s - s_2)^2 + \delta^2 Q_3(s), \qquad \Delta s = s_2 - s_1 > 0,$$

where Q_3 is real analytic in a neighbourhood of s_1, s_2. Assume

$$Q_3(s_1) < 0 < Q_3(s_2).$$

Define

$$\beta = \frac{\sqrt{Q_3(s_2)}}{\Delta s}, \qquad \gamma = \frac{\sqrt{-Q_3(s_1)}}{\Delta s}. \tag{36}$$

Then there is $\delta_0 > 0$ such that, for $0 < \delta < \delta_0$, the four roots of $Q(\,\cdot\,,\delta)$ split into one complex-conjugate pair near s_1 and one real pair near s_2. More precisely, after labelling,

$$r_{1,2}(\delta) = m_1(\delta) \pm \mathrm{i}h_1(\delta), \qquad m_1(\delta) = s_1 + \alpha_1\delta^2 + \mathcal{O}(\delta^4), \qquad h_1(\delta) = \gamma\delta + \mathcal{O}(\delta^3), \tag{37}$$

$$r_{3,4}(\delta) = m_2(\delta) \mp h_2(\delta), \qquad m_2(\delta) = s_2 + \alpha_2\delta^2 + \mathcal{O}(\delta^4), \qquad h_2(\delta) = \beta\delta + \mathcal{O}(\delta^3), \tag{38}$$

where

$$\alpha_1 = \frac{Q_3'(s_1)}{2\Delta s^2} + \frac{Q_3(s_1)}{\Delta s^3}, \qquad \alpha_2 = \frac{Q_3'(s_2)}{2\Delta s^2} - \frac{Q_3(s_2)}{\Delta s^3}. \tag{39}$$

The estimates are uniform for (s_1, s_2) in any compact set on which Δs is bounded away from zero and $Q_3(s_1) \le -\eta < 0 < Q_3(s_2)$.

Proof. We give the argument near s_1; the proof near s_2 is identical. Put $s = s_1 + \delta y$. Then $Q(s,\delta) = 0$ is equivalent, after division by δ^2, to

$$F_1(y,\delta) = -y^2(\delta y - \Delta s)^2 + Q_3(s_1 + \delta y) = 0.$$

At $\delta = 0$,

$$F_1(y,0) = -\Delta s^2 y^2 + Q_3(s_1),$$

whose two simple roots are $y = \pm\mathrm{i}\sqrt{-Q_3(s_1)}/\Delta s = \pm\mathrm{i}\gamma$. Since

$$\partial_y F_1(\pm\mathrm{i}\gamma, 0) = -2\Delta s^2(\pm\mathrm{i}\gamma) \neq 0,$$

the analytic implicit function theorem gives two analytic branches $y_\pm(\delta)$ and hence two roots $s_1 + \delta y_\pm(\delta)$.

To obtain the coefficient of the centre, write $s = s_1 + \rho\delta + \eta\delta^2 + \mathcal{O}(\delta^3)$. Substitution in $Q(s,\delta) = 0$ gives, at order δ^2,

$$\rho^2 = \frac{Q_3(s_1)}{\Delta s^2},$$

and, at order δ^3,

$$-2\rho\eta\Delta s^2 + 2\rho^3\Delta s + Q_3'(s_1)\rho = 0.$$

Thus

$$\eta = \frac{Q_3(s_1)}{\Delta s^3} + \frac{Q_3'(s_1)}{2\Delta s^2} = \alpha_1,$$

independent of the sign of ρ. Therefore the two roots have the same δ^2 centre displacement and opposite imaginary linear splitting.

Near s_2, put $s = s_2 + \delta y$. The rescaled equation is

$$F_2(y,\delta) = -y^2(\Delta s + \delta y)^2 + Q_3(s_2 + \delta y) = 0.$$

At $\delta = 0$ its two simple roots are $y = \pm\sqrt{Q_3(s_2)}/\Delta s = \pm\beta$, and the same calculation gives

$$\eta = \frac{Q_3'(s_2)}{2\Delta s^2} - \frac{Q_3(s_2)}{\Delta s^3} = \alpha_2.$$

Because the coefficients of Q are real and depend on δ only through δ^2, the root centres are even functions of δ and the half-separations are δ times even functions of δ. This gives the displayed $\mathcal{O}(\delta^4)$ and $\mathcal{O}(\delta^3)$ remainders.

Uniformity follows from the same implicit-function argument on a compact parameter set: the derivatives $\partial_y F_j$ at the four limiting simple roots are bounded away from zero, and the relevant derivatives of Q_3 are uniformly bounded. □

Proposition 8.13 (Cross-ratio asymptotic from local branch splitting)**.** *Assume the hypotheses of lemma 8.12, and order the roots as*

$$r_1 = m_1 + \mathrm{i}h_1, \qquad r_2 = m_1 - \mathrm{i}h_1, \qquad r_3 = m_2 - h_2, \qquad r_4 = m_2 + h_2.$$

Then the cross-ratio

$$\mathrm{CR} = \frac{(r_1 - r_3)(r_2 - r_4)}{(r_1 - r_4)(r_2 - r_3)}$$

satisfies

$$\mathrm{CR} = 1 - \frac{4\mathrm{i}\,\beta\gamma}{\Delta s^2}\,\delta^2 + \mathcal{O}(\delta^4), \qquad \arg(\mathrm{CR}) = -\frac{4\beta\gamma}{\Delta s^2}\,\delta^2 + \mathcal{O}(\delta^4). \tag{40}$$

Consequently, along any high-fold branch with $\delta_k^2 = A^2/k + \mathcal{O}(k^{-2})$ *and* $s_{2,k} = s_{2,\infty} + \mathcal{O}(k^{-1})$,

$$\arg(\mathrm{CR}_k)\,k = -4\,\frac{A^2\beta_\infty\gamma_\infty}{\Delta s_\infty^2} + \mathcal{O}(k^{-1}).$$

In particular, after proposition 8.15, this yields $\arg(\mathrm{CR}_k)k = -4 + \mathcal{O}(k^{-1})$ *on the local high-fold branch.*

Proof. Let $D = m_2 - m_1$ and set

$$z = \frac{h_2 + \mathrm{i}h_1}{D}.$$

Using the root ordering above,

$$(r_1 - r_3)(r_2 - r_4) = D^2 - (h_2 + \mathrm{i}h_1)^2,$$
$$(r_1 - r_4)(r_2 - r_3) = D^2 - (h_2 - \mathrm{i}h_1)^2.$$

Hence

$$\mathrm{CR} = \frac{1 - z^2}{1 - \bar{z}^2}.$$

By lemma 8.12,

$$D = \Delta s + \mathcal{O}(\delta^2), \qquad h_1 = \gamma\delta + \mathcal{O}(\delta^3), \qquad h_2 = \beta\delta + \mathcal{O}(\delta^3),$$

and therefore

$$z = \frac{\delta(\beta + \mathrm{i}\gamma)}{\Delta s} + \mathcal{O}(\delta^3).$$

Expanding the exact expression gives

$$\mathrm{CR} = 1 + (\bar{z}^2 - z^2) + \mathcal{O}(\delta^4) = 1 - \frac{4\mathrm{i}\beta\gamma}{\Delta s^2}\delta^2 + \mathcal{O}(\delta^4).$$

Since the leading correction is purely imaginary, the same leading coefficient gives the argument. The final high-fold statement follows by substituting $\delta_k^2 = A^2/k + \mathcal{O}(k^{-2})$ and $s_{2,k} = s_{2,\infty} + \mathcal{O}(k^{-1})$, so that $\Delta s, \beta, \gamma$ may be replaced by their limiting values at the cost of an additional $\mathcal{O}(k^{-1})$ after multiplication by k. □

Lemma 8.14 (Limiting axial equation in polynomial form)**.** *Fix* s_1 *and let* $c > s_1$ *be a candidate pinching centre. Put*

$$\Delta_c = c - s_1, \qquad P_c = U^{-2}(2(s_1 + c)U_1' - U_2), \qquad M_c = U^{-2}(U' + s_1 c\,U_1'),$$

and

$$\tilde{Q}_{2,c}(x) = P_c x^2 - 2M_c x - 1.$$

Let $X_0(c)$ denote the leading reduced axial coefficient from eq. (72). Then

$$X_0(c) = \frac{\pi U^2}{\Delta_c^3}\,\Xi(c), \qquad \Xi(c) := P_c\, s_1 c - M_c(s_1 + c) - 1. \tag{41}$$

Consequently, the limiting axial condition $X_0(C_2) = 0$ is equivalent to

$$\Xi(C_2) = P_{C_2} s_1 C_2 - M_{C_2}(s_1 + C_2) - 1 = 0. \tag{42}$$

Proof. Write $q = Q_3(c)$, $q' = Q_3'(c)$ and $\Delta = \Delta_c$. The branch-point expansion from lemma C.2 gives the real-oval midpoint $c + \alpha\delta^2 + \mathcal{O}(\delta^4)$, where

$$\alpha = \frac{q'}{2\Delta^2} - \frac{q}{\Delta^3}.$$

Using the desingularized coordinate

$$x = c + \delta\frac{\sqrt{q}}{\Delta}\sin\varphi + \alpha\delta^2 + \mathcal{O}(\delta^3),$$

and expanding the axial numerator $Q_2(x) = -(x - s_1)(x - c) + \delta^2 U_1' x$, the odd order-$\delta$ term integrates to zero on the symmetric φ-interval. The first non-zero term is

$$F_{\mathrm{ax}}(\delta, c) = \pi\delta^2\left(\frac{U_1' c}{\Delta} - \frac{q'}{2\Delta^2} + \frac{q}{\Delta^3}\right) + \mathcal{O}(\delta^4).$$

The parity argument in lemma C.4 excludes an order-δ^3 contribution. Since

$$q = 2U_1' c^3 - U_2 c^2 - 2U' c - U^2, \qquad q' = 6U_1' c^2 - 2U_2 c - 2U',$$

a direct simplification gives

$$U_1' c\,\Delta^2 - \frac{\Delta q'}{2} + q = U^2(P_c s_1 c - M_c(s_1 + c) - 1).$$

Dividing by Δ^3 yields (41). □

Proposition 8.15 (Closure identity for the cross-ratio coefficient)**.** *Let C_2 be the local high-fold pinching centre from theorem C.6. Assume the sign and separation conditions*

$$\Delta_\infty := C_2 - s_1 > 0, \qquad Q_3(s_1) < 0 < Q_3(C_2),$$

and define

$$\beta_\infty = \frac{\sqrt{Q_3(C_2)}}{\Delta_\infty}, \qquad \gamma_\infty = \frac{\sqrt{-Q_3(s_1)}}{\Delta_\infty}.$$

Then the amplitude from eq. (80) satisfies the exact identity

$$\frac{A^2\beta_\infty\gamma_\infty}{\Delta_\infty^2} = 1. \tag{43}$$

Consequently, on the local high-fold branch of theorem C.6, the cross-ratio obeys

$$\arg(\mathrm{CR}_k)\, k = -4 + \mathcal{O}(k^{-1}) \tag{44}$$

under the root ordering of proposition 8.13.

Proof. For brevity write

$$C = C_2, \quad \Delta = C - s_1, \quad P = P_C, \quad M = M_C, \quad T(x) = \tilde{Q}_{2,C}(x) = Px^2 - 2Mx - 1.$$

By lemma 8.14, the limiting axial condition is

$$\Xi := Ps_1C - M(s_1 + C) - 1 = 0.$$

Two elementary identities for the quadratic T now close the calculation. First,

$$T(s_1)T(C) + (P + M^2)\Delta^2 = \big(Ps_1C - M(s_1 + C) - 1\big)^2 = \Xi^2. \tag{45}$$

Since $Z_0^2 = P + M^2$ and $\Xi = 0$, this gives

$$T(s_1)T(C) = -Z_0^2\Delta^2. \tag{46}$$

Second,

$$\Delta\, T'(C) - 2T(C) = -2\Xi = 0, \qquad \text{hence} \qquad T'(C) = \frac{2T(C)}{\Delta}. \tag{47}$$

The endpoint evaluations of T are exactly the cubic spectral polynomial divided by U^2:

$$T(s_1) = \frac{Q_3(s_1)}{U^2}, \qquad T(C) = \frac{Q_3(C)}{U^2}. \tag{48}$$

Using the sign assumptions and (46), the positive square roots satisfy

$$\sqrt{-Q_3(s_1)}\,\sqrt{Q_3(C)} = U^2 Z_0 \Delta. \tag{49}$$

Moreover, substituting (47) into the amplitude formula (80) gives

$$A^2 = \frac{2T(C)^2\Delta^2}{Z_0 T'(C) Q_3(C)} = \frac{T(C)\Delta^3}{Z_0 Q_3(C)} = \frac{\Delta^3}{Z_0 U^2},$$

where the last equality uses $T(C) = Q_3(C)/U^2$. Therefore

$$\frac{A^2\beta_\infty\gamma_\infty}{\Delta^2} = \frac{\Delta^3}{Z_0U^2}\,\frac{\sqrt{Q_3(C)}\sqrt{-Q_3(s_1)}}{\Delta^2}\,\frac{1}{\Delta^2} = \frac{\sqrt{Q_3(C)}\sqrt{-Q_3(s_1)}}{Z_0U^2\Delta} = 1$$

by (49). Finally, (44) follows by substituting (43) into proposition 8.13. □

Remark 8.16 (Sign convention for arg(CR)). The sign of $\arg(\mathrm{CR}) \cdot k \to -4$ depends on the ordering of the real roots r_3, r_4 in the cross-ratio. Swapping $r_3 \leftrightarrow r_4$ replaces CR by $1/\mathrm{CR} = \overline{\mathrm{CR}}$ (since $|\mathrm{CR}| = 1$), giving $\arg(\mathrm{CR}) \cdot k \to +4$. Thus the *intrinsic* statement is $|\arg(\mathrm{CR})| \cdot k \to 4$; the sign $\pm$ records only the branch-point labelling convention. Throughout this paper we use $r_3 < r_4$, yielding -4.

Remark 8.17 (Universality of the factor 4). The value -4 in (35) is *not* tied to $\deg(Q) = 4$ or to $2(\chi - 2)$ individually. It is a *universal* consequence of the cross-ratio formula applied to any four points that coalesce into two symmetric pairs of mixed type (one real, one complex conjugate). For a degree-6 spectral polynomial with three coalescing pairs, one would still select four of the six roots for the cross-ratio, and the same factor 4 would appear. The coincidence $\deg(Q) = 4$ with the cross-ratio factor is just that: a coincidence.

Remark 8.18 (Interpretation). The gauge flow $s_1 \mapsto (\delta(s_1), s_2(s_1))$ is an *isospectral* flow: it moves a divisor on a fixed spectral curve $\mu^2 = Q(s)$ whose conformal type (measured by CR) is preserved at ω_{crit}. This is distinct from Painlevé VI isomonodromy, where the cross-ratio itself is the independent variable. Here τ is fixed and the cross-ratio is a *numerically conserved quantity* of the particular flow, not the deformation parameter.

The value -4 in (35) was previously attributed to either $-\deg(Q)$ or $2(\chi-2)$; proposition 8.13 identifies the *structural factor* 4 in the cross-ratio expansion under pairwise root coalescence, and proposition 8.15 proves the closure identity that fixes the exact coefficient. Genus-2 data are therefore not needed for the explanation.

Barrier and monodromy remarks.

Remark 8.19 (Two independent obstructions beyond genus 1). Extending the BHS framework to higher genus encounters *two independent barriers*.
(i) Topological obstruction ($g_{\text{surf}} \geq 2$). The Bobenko–Eitner divisor formula [12] gives $\deg(D_u) = 4g_{\text{surf}} - 4$ for the umbilical divisor of an isothermic surface of genus g_{surf}. For $g_{\text{surf}} = 1$: $\deg(D_u) = 0$, so the surface is generically umbilic-free and supports a *global* conformal-curvature-line parametrization; this is precisely the setting exploited by [1, 2]. For $g_{\text{surf}} = 2$: $\deg(D_u) = 4$, so four umbilics are topologically unavoidable, destroying the conformal-line net.
(ii) Spectral rigidity ($g_{\text{spectral}} = 1$ forced). The Darboux classification [9] shows that every isothermic surface with one family of planar curvature lines satisfies the Riccati equation $h_u = U(u)\,\mathrm{e}^{h} + U_1(u)\,\mathrm{e}^{-h}$ [2, Theorem 1]. Compatibility with harmonicity ($h_{uu} + h_{ww} = 0$) forces U, U_1 to be the two linearly independent solutions of the integer Lamé equation with one-gap potential [2, Lemma 1]: $y'' = (C_1 - 8UU_1)\,y$. The Corollary 1 identity of [2] then gives h_w^2 as a degree-4 polynomial in e^{-h}. At the critical point ω satisfying $\vartheta_2'(\omega) = 0$, the highest-order term in that quartic model vanishes, leaving the cubic (11) as the effective spectral polynomial at the critical slice. This forces the cubic (11) and hence $\deg Q \leq 4$, so the spectral curve $\mu^2 = Q(s)$ has genus at most 1. Raising the spectral genus to 2 would require replacing the two-term Riccati equation with a higher-order nonlinear ODE—i.e., leaving the class of isothermic surfaces with planar curvature lines entirely.

The two barriers are *logically independent*: (i) obstructs $g_{\text{surf}} \geq 2$, while (ii) constrains $g_{\text{spectral}} \leq 1$ even for tori ($g_{\text{surf}} = 1$). Together they imply that compact non-CMC Bonnet pairs with spectral genus > 1 cannot exist within the BHS framework. The Wente tori [16]—which *do* possess genus-2 spectral curves—are CMC and arise only as a limit in which the BHS spectral curve degenerates and is replaced by a sinh-Gordon spectral curve of a fundamentally different type [18]. Note that proposition 8.13 provides a structural explanation for the value -4 independently of genus-2 data: the factor is structural, arising from pairwise root coalescence rather than from $\deg(Q)$ or χ.

Remark 8.20 (Spectral-curve pinch in the tested CMC-direction sweeps). The spectral rigidity of barrier (ii) in remark 8.19 can be probed numerically by measuring how far each BHS torus is from being a CMC surface. We define the *relative CMC deviation*

$$\rho \;=\; \frac{\sigma(H)}{|\bar{H}|}, \qquad \sigma(H)^2 = \frac{1}{|\Sigma|}\int_{\Sigma}(H - \bar{H})^2\, dA,$$

where $\bar{H}$ is the area-averaged mean curvature. For a genuine CMC surface $\rho = 0$; for a Delaunay unduloid the discrete analogue satisfies $\rho < 10^{-3}$ at our grid resolution.

Computing ρ on BHS isothermic tori for every validated seed $k = 3, \ldots, 12$ gives $\rho \in [0.63,\, 1.45]$—firmly $\mathcal{O}(1)$—with a minimum at $k = 6$ and no monotone trend. To test whether a simple one-parameter rescaling of the Bonnet data can move toward the CMC condition, we fix $k = 5$, $\tau_{\text{im}} = 0.3205$, $s_1 = -3.80$ and rescale $(\delta, s_2) \mapsto \lambda(\delta_0, s_{2,0})$ for $\lambda \in (0, 1]$. As λ decreases from 1, ρ first reaches a shallow minimum ($\rho = 0.61$ at $\lambda \approx 0.78$), then *diverges* ($\rho \approx 22$ at $\lambda = 0.29$). At $\lambda \approx 0.22$ the quartic $Q(s)$ in Corollary 1 of [2] loses its two real roots—they collide and escape into the complex plane—so the real oval of the spectral curve pinches off and the integration domain ceases to exist. An independent sweep $\tau_{\text{im}} \to \tau_{\text{crit}} \approx 0.3547$ (where $\omega \to 0$) shows ρ oscillating chaotically in $[1.1,\, 61]$ with no discernible trend ($R^2 < 0.01$ for every power-law candidate).

These experiments suggest that, along the tested rescaling and τ-deformation directions, the BHS family does not approach a CMC limit smoothly before the real oval disappears. In that restricted numerical sense, the spectral pinch behaves like a barrier in moduli space. This is consistent with the Riccati/Lamé obstruction forcing $g_{\text{spectral}} \leq 1$, but it is not a theorem about all possible deformation directions.

Remark 8.21 (Monodromy structure of PVI with Manin parameters). The Manin parameters $[\alpha, \beta, \gamma, \delta] = [0, 0, 0, 0]$ force all four Fuchsian monodromy matrices $M_j \in \mathrm{SL}(2, \mathbb{C})$ ($j \in \{0, t, 1, \infty\}$) to be *unipotent* in the sense that $\mathrm{tr}(M_j) = 2$; in nontrivial cases this corresponds to a Jordan block of size 2. The closure relation $M_0 M_t M_1 M_\infty = I$ restricts the gauge-invariant data to the Fricke coordinates $p_k = \mathrm{tr}(M_i M_j)$ (where $\{i, j, k\} = \{0, t, 1\}$), which satisfy the Cayley cubic

$$p_1^2 + p_2^2 + p_3^2 + p_1 p_2 p_3 - 4(p_1 + p_2 + p_3) + 8 = 0. \tag{50}$$

An explicit unipotent representative consistent with (50) is [13]:

$$M_\infty = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \; M_0 = \begin{pmatrix} 1 & 0 \\ -x_1^2 & 1 \end{pmatrix}, \; M_1 = \begin{pmatrix} 1 - x_1 x_2 x_3 & x_2^2 \\ -x_3^2 & 1 + x_1 x_2 x_3 \end{pmatrix},$$

with $p_k = 2 - x_k^2$ and $M_t = M_0^{-1} M_1^{-1} M_\infty^{-1}$. The standard Jimbo connection parameter σ_{0t} satisfies $\mathrm{tr}(M_0 M_t) = 2\cos(\pi\sigma_{0t})$; thus the logarithmic regime occurs precisely when the relevant monodromy data satisfy $\mathrm{tr}(M_0 M_t) = 2$, i.e. $\sigma_{0t} = 0$. In that case the Jimbo (1982) asymptotic formula degenerates to a *logarithmic* regime [13]:

$$y(t) \sim \frac{t}{\ln t + C_0}(1 + \mathcal{O}(t)) \qquad (t \to 0), \tag{51}$$

where the constant C_0 replaces σ as the effective connection parameter. The non-trivial monodromy data is thus encoded in the Stokes-like parameters (x_1, x_2, x_3) on the Cayley cubic, not in the standard connection exponent.

Spectral modulus, corrected lambda transform, and finite-data audit.

Conjecture 8.22 (Spectral modular parameter)**.** *Let $\tau_{\mathrm{spec}}(k)$ denote the modular parameter of the spectral curve $w^2 = Q(s)$ at $\omega = \omega_{\mathrm{crit}}$. Then*

$$\tau_{\mathrm{spec}}(k) \;=\; -\frac{1}{2} + \mathrm{i}\,\frac{\ln(4k)}{\pi} + \mathcal{O}(k^{-2}), \tag{52}$$

supported numerically at the 6-significant-digit level on 50 archived spectral-period seeds spanning $k = 5$ to 200. The residual $|\tau_{\mathrm{spec}}^{\mathrm{num}} - \tau_{\mathrm{spec}}^{\mathrm{formula}}| < 3 \times 10^{-7}$ for the tested seeds with $k \geq 50$.

Remark 8.23. The real part $\mathrm{Re}\,\tau_{\mathrm{spec}} = -\frac{1}{2}$ is accounted for in conditional theorem 8.27 under the corrected branch-normalization assumptions. The imaginary part $\mathrm{Im}\,\tau_{\mathrm{spec}} = \ln(4k)/\pi$ rests entirely on numerical evidence (6-digit agreement) and remains open.

Observation 8.24 (Spectral nome and corrected lambda transform)**.** Assuming conjecture 8.22, the Jacobi nome of the spectral curve takes the leading-order form

$$q_{\mathrm{spec}}(k) \;:=\; \mathrm{e}^{\mathrm{i}\pi\tau_{\mathrm{spec}}} \;\sim\; \frac{-\mathrm{i}}{4k} \qquad (k \to \infty). \tag{53}$$

With the mixed branch-point ordering used in (31), the elliptic modular lambda value is not itself the cross-ratio. The corrected branch relation is

$$\mathrm{CR}_k \;=\; \frac{1}{1 - \lambda(\tau_{\mathrm{spec}}(k))}. \tag{54}$$

Equivalently,

$$\lambda(\tau_{\mathrm{spec}}(k)) \;=\; 1 - \frac{1}{\mathrm{CR}_k}. \tag{55}$$

On the archived full-precision spectral-period seed subset consisting of 50 checked seeds in the range $k = 5$ to 200, a high-precision seed-level audit of this branch relation gives errors $< 10^{-11}$.

On the rounded printed high-k seed values, the same audit gives errors of order 10^{-10}, consistent with the printed rounding. For larger archived spectral-period seeds, the period computation remains numerically more delicate, so we do not promote the finite-data claim beyond the verified $k \le 200$ range here. Since $|\mathrm{CR}_k| = 1$ (lemma 8.9) and $\arg(\mathrm{CR}_k) \cdot k \to -4$ (proposition 8.13), the cross-ratio admits the asymptotic approximation

$$\mathrm{CR}_k \sim \mathrm{e}^{-4\mathrm{i}/k} \qquad (k \to \infty). \tag{56}$$

Remark 8.25. Equation (56) is an asymptotic statement, not an exact identity: at finite k the sub-leading corrections in (61) contribute to both q_{spec} and $\lambda(\tau_{\mathrm{spec}})$. The corrected finite branch transform is (54). The q-expansion $\lambda(\tau) = 16q - 128q^2 + 704q^3 + \mathcal{O}(q^4)$ gives, from $q_{\mathrm{spec}} \sim -\mathrm{i}/(4k)$,

$$\lambda(\tau_{\mathrm{spec}}) = -\frac{4\mathrm{i}}{k} + \mathcal{O}(k^{-2}), \qquad \frac{1}{1-\lambda(\tau_{\mathrm{spec}})} = 1 - \frac{4\mathrm{i}}{k} + \mathcal{O}(k^{-2}),$$

which matches $\mathrm{CR}_k \sim \mathrm{e}^{-4\mathrm{i}/k}$. Among the six standard anharmonic transforms, this is the unique one that is finite, tends to 1, and has first-order argument $-4/k$ under the present sign convention.

Lemma 8.26 (Lambda boundary circle at the $q = -\mathrm{i}\rho$ cusp)**.** *Let $\lambda(\tau)$ denote the modular lambda function and put $q = \mathrm{e}^{\pi\mathrm{i}\tau}$. For every $y > 0$,*

$$\tau = -\frac{1}{2} + \mathrm{i}y \qquad \Longrightarrow \qquad |1 - \lambda(\tau)| = 1. \tag{57}$$

Moreover, for each sufficiently large y this boundary component is locally unique: there is a neighbourhood $|x + \frac{1}{2}| < \eta(y)$ such that

$$|1 - \lambda(x + \mathrm{i}y)| = 1 \quad \Longrightarrow \quad x = -\frac{1}{2}. \tag{58}$$

Proof. The modular lambda function satisfies

$$\lambda(\tau + 1) = \frac{\lambda(\tau)}{\lambda(\tau) - 1}. \tag{59}$$

If $\tau = -\frac{1}{2} + \mathrm{i}y$, then $\tau + 1 = -\overline{\tau}$. Since the q-expansion of λ has real coefficients, $\lambda(\tau+1) = \overline{\lambda(\tau)}$. Writing $z = \lambda(\tau)$, this gives

$$\bar{z} = \frac{z}{z-1}.$$

Equivalently, $|z|^2 = 2\,\mathrm{Re}\, z$, and hence

$$|1 - z|^2 = 1 - 2\,\mathrm{Re}\, z + |z|^2 = 1.$$

This proves (57).

For the local uniqueness statement, set

$$G(x, y) = |1 - \lambda(x + \mathrm{i}y)|^2 - 1.$$

Near the cusp $q = 0$,

$$\lambda(\tau) = 16q - 128q^2 + \mathcal{O}(q^3), \qquad q = \rho \mathrm{e}^{\pi\mathrm{i}x}, \quad \rho = \mathrm{e}^{-\pi y}.$$

Therefore

$$G(x, y) = -32\rho\cos(\pi x) + \mathcal{O}(\rho^2), \qquad \partial_x G(x, y) = 32\pi\rho\sin(\pi x) + \mathcal{O}(\rho^2).$$

At $x = -\frac{1}{2}$ the derivative is $-32\pi\rho + \mathcal{O}(\rho^2)$, which is non-zero for y large enough. By the implicit function theorem the zero set of G in a small neighbourhood of $x = -\frac{1}{2}$ is locally a unique graph. The neighbourhood may depend on y; a uniform strip for all $y \ge Y$ would require an explicit uniform bound. Since the whole line $x = -\frac{1}{2}$ already lies in this zero set by (57), that graph is exactly $x = -\frac{1}{2}$. □

Conditional Theorem 8.27 (Real part of the spectral modulus under the corrected lambda branch)**.** *Under the following hypotheses:*

(H1) *the mixed root topology and ordering $r_1 = c$, $r_2 = \bar{c}$, $r_3 = a$, $r_4 = b$ holds, established either asymptotically by theorem 8.5, from the exact finite certificates summarised in observation 8.8, or from the interval-certificate hypotheses of proposition 8.7;*

(H2) *the corrected branch relation (54) applies, namely $\mathrm{CR}_k = (1 - \lambda(\tau_{\mathrm{spec}}(k)))^{-1}$;*

(H3) *the spectral modulus is taken in the high-cusp branch whose nome satisfies*

$$q_{\mathrm{spec}} = \mathrm{e}^{\pi \mathrm{i} \tau_{\mathrm{spec}}} \sim -\frac{\mathrm{i}}{4k},$$

so that, for k large enough, τ_{spec} lies in the local neighbourhood governed by lemma 8.26, for all sufficiently large fold numbers k on that branch,

$$\mathrm{Re}\ \tau_{\mathrm{spec}}(k) = -\frac{1}{2}. \tag{60}$$

Equivalently, the same statement holds modulo $\mathrm{SL}_2(\mathbb{Z})$ after choosing this cusp representative.

Proof. By (H1), the mixed root ordering gives $|\mathrm{CR}_k| = 1$ by lemma 8.9. Using the corrected transform (54) from (H2), this is equivalent to

$$|1 - \lambda(\tau_{\mathrm{spec}}(k))| = 1.$$

The nome condition $q_{\mathrm{spec}} \sim -\mathrm{i}/(4k)$ from (H3) selects the cusp component with $\mathrm{Re}\,\tau$ near $-\frac{1}{2}$. The local uniqueness part of lemma 8.26 then forces $\mathrm{Re}\,\tau_{\mathrm{spec}} = -\frac{1}{2}$ on this branch for k large enough. □

Observation 8.28 (Universality of CR_k on the tested τ_{BHS} sweeps)**.** Numerically, on the tested τ-deformation sweeps, the cross-ratio CR_k behaves as a function of k alone: no dependence on τ_{BHS} is resolved within numerical precision. For $k \in \{5, 10, 20\}$ and $\mathrm{Im}\ \tau_{\mathrm{BHS}} \in [0.20, 0.35]$ (97 stored points from the τ-deformation sweep: 41 for $k = 5$ and 28 each for $k = 10, 20$), $|\mathrm{CR}| = 1$ to 12 digits and CR_k is constant in τ_{BHS} to machine precision. This universality is non-trivial: the solution $\delta_k(\tau_{\mathrm{BHS}})$ of the periodicity conditions depends on τ_{BHS} through the theta-function coefficients (the amplitude A varies from 6.78 to 7.27 across the tested range; validated claim 6.1), so the cancellation in the cross-ratio is strong numerical evidence for a structural property.

Observation 8.29 (Asymptotic refinement of τ_{spec})**.** Assuming conjecture 8.22, the formula is the leading term of an asymptotic expansion. Numerically, a secondary empirical fit on the tested spectral-period seeds is consistent with

$$\tau_{\mathrm{spec}}(k) \ = \ -\frac{1}{2} + \frac{\mathrm{i}}{\pi}\left[\ln(4k) - \frac{1}{8k^2} + \mathcal{O}(k^{-4})\right]. \tag{61}$$

Equivalently, defining the residual $R(k) := \mathrm{Im}(\tau_{\mathrm{spec}}) \cdot \pi / \ln(4k) - 1$, one finds $R(k) \cdot k^2 \cdot \ln(4k) \to -1/8$ with five stable digits for $k \leq 100$. The next correction is $\mathcal{O}(k^{-4})$ with estimated coefficient $\approx -0.039/\pi$. This should be read only as a numerical refinement of the conjectural leading term, not as a proved or independently certified higher-order law.

Remark 8.30 (Interpretation of τ_{spec}). The real part $\mathrm{Re}\ \tau_{\mathrm{spec}} = -1/2$ is supported by the spectral-period computations and is accounted for by conditional theorem 8.27 once the corrected anharmonic transform between $\lambda(\tau_{\mathrm{spec}})$ and the branch-point cross-ratio is fixed. The imaginary part $\mathrm{Im}\ \tau_{\mathrm{spec}} = \ln(4k)/\pi - 1/(8\pi k^2) + \mathcal{O}(k^{-4})$ says the spectral curve degenerates logarithmically as $k \to \infty$, with a clean sub-leading correction. Formula (52) is best read at present as a conjectural organizing relation between the discrete fold number k and the conformal type of the spectral curve (OP3 below), not yet as a proved bridge.

Conjecture 8.31 (Connection constant C_0)**.** *The logarithmic connection constant C_0 appearing in (51) satisfies*

$$C_0 \;=\; \ln\!\Big(\frac{A^2}{2\pi}\Big) + \gamma_{\mathrm{EM}} \;\approx\; 2.654\,287, \tag{62}$$

where $A = \lim_{k\to\infty} \delta\sqrt{k}$ is the amplitude constant of eq. (27), γ_{EM} is the Euler–Mascheroni constant, and the numerical value uses the Richardson-extrapolated $A^2 \approx 50.1465$ (eq. (21)).

Remark 8.32. The conjectured form (62) is motivated by the structure of Jimbo's asymptotic formula in the unipotent case [13]; however, direct numerical extraction of C_0 from $y(t;\sigma_{0t}=0)$ is frustrated by the slow logarithmic convergence in (51). The *correct* Hamiltonian form for PVI(0, 0, 0, 0) that was essential for the initial conditions reads

$$t(t-1)\,H \;=\; y(y-1)(y-t)\,p^2 - \big[\theta_0\,y(y-1) + \theta_1\,y(y-t)\big]\,p,$$

with $\theta_0 = \theta_1 = 0$; the fourth exponent $\theta_\infty = 1$ from the Kitaev double folding [7] gives $\delta = (1-\theta_\infty^2)/2 = 0$ in the DLMF convention. This reduces the Hamilton–Jacobi problem to two quadratures but does not eliminate the ε-dependence that prevents extraction of C_0.

8.4 Numerical topology of the moduli space

We now summarize the numerically observed structure of $\mathcal{M}$ at fixed τ. Write γ_k for the moduli curve at fold number k, viewed as a curve in the (δ, s_2)-plane parametrized by s_1.

Computational Certificate 8.33 (Simple-arc embedding on the tested folds)**.** *For every tested fold number $k \in \{5,\dots,24,\ 25,\ 50,\ 100\}$, the curve γ_k is a simple arc in the (δ, s_2)-plane. Specifically:*

(i) *For $k = 5,\dots,21$ and $k = 25, 50, 100$, both $\delta(s_1)$ and $s_2(s_1)$ are strictly monotone.*

(ii) *For $k = 22, 23, 24$, $s_2(s_1)$ is strictly monotone but $\delta(s_1)$ exhibits a small non-monotone bump near the fold boundary ($s_1 \approx -2.75$); nevertheless, "injectivity" of s_2 prevents self-intersection, and γ_k remains a simple arc. The bump amplitude decays geometrically:* 0.0156 *($k = 22$),* 0.0079 *($k = 23$),* 0.0032 *($k = 24$), and vanishes by $k = 25$.*

(iii) *For $k = 7$, two numerically distinct continuation starts collapse onto the same curve throughout the tracked range: after branch matching, the two datasets agree to $O(10^{-11})$ and no separate branch survives.*

For $k = 8,\dots,24, 25, 50, 100$, the archived extended sweeps cover $s_1 \in [-10.0,\ -1.75]$ with step 0.25*, producing more than* 600 *validated solutions in total, all satisfying the three Lemma 8 checks. For $k = 5, 6, 7$, the corresponding low-fold data are taken from the separate archived continuations on* $[-10.0,\ -2.75]$.

Observation 8.34 (998-seed cross-section)**.** At the reference gauge $s_1 = -8.5$, the 998 seeds $k = 3,\dots,1000$ yield 998 distinct points (δ_k, s_{2k}) on $\mathcal{M}$. All residuals satisfy $\|F(s_1,\delta_k,s_{2k})\| < 10^{-9}$. The map $k \mapsto (\delta_k, s_{2k})$ is injective with δ strictly decreasing and s_2 strictly increasing — hence the cross-section is a simple arc.

Remark 8.35 (Transition and nesting). For $k = 22, 23, 24$, the map $s_1 \mapsto \delta(s_1)$ develops a turning point at $s_1 = -2.75$: δ decreases from $s_1 = -10$ to $s_1 = -2.75$, then increases slightly before the fold boundary. The bump decays rapidly (from 1.56×10^{-2} at $k = 22$ to 3.22×10^{-3} at $k = 24$) and disappears by $k = 25$. Despite the non-monotonicity in δ, $s_2(s_1)$ remains strictly decreasing, so γ_k has no self-intersection. The curves $\gamma_5, \gamma_6, \gamma_7$ are nested: at every tested $s_1 \in [-10,\ -2.75]$, the inter-fold separations $\Delta\delta_{k,k+1}$ are positive and decreasing in k, matching the asymptotic $\Delta\delta \sim A/(2k^{3/2})$ from section 6.4.

Table 3: Selected values of (δ, s_2) along the transversal $\mathcal{T}$ at $s_1 = -8.5$. The product $\delta^2 k$ converges to $A^2 \approx 50.1$ (section 6).

k	δ	s_2	$\delta^2 k$
3	3.20159	2.5061	30.75
5	2.70793	3.8474	36.66
10	2.06311	5.3040	42.56
50	0.98458	6.8647	48.47
100	0.70212	7.0902	49.30
500	0.31615	7.2760	49.97
1000	0.22374	7.2996	50.06

Observation 8.36 (Fold boundary refinement)**.** Linear extrapolation of $1/\det(J)$ along each γ_k refines the fold locations of observation 8.4: $s_1^*(5) \approx -2.27$, $s_1^*(6) \approx -1.94$, $s_1^*(7) \approx -1.63$. As $s_1 \to s_1^*$, the dominant singular direction of the Jacobian becomes increasingly pronounced in the numerical diagnostics. The derivative $d\delta/ds_1$ accelerates toward the boundary (positive second derivative), consistent with a square-root cusp in $\delta(s_1)$ at s_1^*.

Observation 8.37 (Tested slice numerically consistent with contractibility)**.** On the tested slice $\tau = \frac{1}{2} + \frac{25}{78}\, i$, the archived data are numerically consistent with contractibility: each γ_k is an embedded arc terminating at the fold $s_1^*(k)$, the transversal cross-section is a simple arc (observation 8.34), and no non-trivial cycles are detected in the archived $k = 5, 6, 7$ sweep data, in the archived extended sweeps for $k = 8, \ldots, 24, 25, 50, 100$ (more than 600 validated points, all Lemma 8 compliant), nor in the 998-seed dataset.

Remark 8.38 (Multi-τ extension of the simple-arc picture)*.* The same simple-arc / no-detected-cycle picture extends beyond the reference lattice τ_{ref} on the tested sweeps. For each τ_{im} in the set $\{0.20, 0.25, 0.30, 0.35\}$, the k-cross-section at $s_1 = -8.5$ ($k = 3, \ldots, 200$) is a simple arc (δ strictly decreasing, s_2 strictly increasing), and the s_1-continuation at $k = 5$ yields an embedded arc (s_2 strictly monotone). At $\tau_{\text{im}} = 0.20$ (far from τ_{ref}), γ_k develops δ-folds near the boundary of its existence region. Extended tests at $k \in \{5, 10, 25, 50, 100\}$ show that the fold amplitude decays as $k^{-0.22}$: from 0.84 at $k = 10$ to 0.40 at $k = 100$. At $k \geq 50$, a second fold appears, but s_2-injectivity is preserved in all cases, so every γ_k remains an embedded arc and the tested slices continue to show no numerical evidence against contractibility.

Bisection in τ at $k = 5$ locates the fold onset at

$$\tau_{\text{fold}} = 0.24475 \pm 2 \times 10^{-6}.$$

For $\tau_{\text{im}} > \tau_{\text{fold}}$, the branches are strictly monotone (simple arcs); for $\tau_{\text{im}} < \tau_{\text{fold}}$, δ-folds are present but s_2-monotonicity is maintained. Together with the saddle-node boundary $\lambda_0 \approx 0.35473$ where ω_{crit} ceases to exist, this yields a hierarchy of two bifurcation points for the lattice geometry: $\tau_{\text{fold}} < \operatorname{Im} \tau_{\text{ref}} < \lambda_0$.

Observation 8.39 (Transversal geometry)**.** The transversal curve $\mathcal{T}\colon k \mapsto (\delta_k, s_{2k})$ at $s_1 = -8.5$ has total arc length 5.82, of which 52% lies in $k = 3, \ldots, 10$ and 9% in $k = 100, \ldots, 1000$. The δ-s_2 correlation is -0.925 (strong anti-correlation), and the tangent angle rotates from 110° ($k = 3$) to 168° ($k = 1000$), indicating progressive alignment with the s_2-axis as $\delta \to 0$.

9 Retraction Form Validation

9.1 The retraction form of Burstall et al.

Let $f\colon \Sigma \to \mathbb{R}^3$ be an isothermic surface and let $x = \sigma^{-1}(f) \in S^3 \subset \mathbb{H}$ be its lift to the 3-sphere via inverse stereographic projection. The *retraction* 1*-form* of [3] is

$$\omega = \frac{x_u}{|x_u|^2}\,\mathrm{d}u - \frac{x_v}{|x_v|^2}\,\mathrm{d}v. \tag{63}$$

The Bonnet pair is then given by

$$\mathrm{d}F^+ = \tfrac{1}{2}\,x\,\bar{\omega}, \qquad \mathrm{d}F^- = -\tfrac{1}{2}\,\bar{\omega}\,x. \tag{64}$$

Remark 9.1. The original formulation in [3] is for isothermic surfaces in S^4. For tori in $\mathbb{R}^3$, we project via $\sigma\colon S^3 \setminus \{p\} \to \mathbb{R}^3$ and verify that all so(4) conditions specialise correctly.

9.2 Implementation with analytic derivatives

Key to the high precision of our validation is the use of *analytic* (not numerical) derivatives. Writing $\psi(z) = \vartheta_1'(z)/\vartheta_1(z)$, we express x_u, x_v, and all required quantities in terms of ψ and its derivatives, which are computed from the theta series (1)–(4).

9.3 Five-gate validation

The retraction form pipeline applies five quality gates to each seed:

R1. Closure ($d\omega = 0$). $\max|\partial_v\omega_u - \partial_u\omega_v|$ normalized by $\max(\|\partial_v\omega_u\|, \|\partial_u\omega_v\|)$. Threshold: < 0.05.

R2. Cross condition ($\bar{\omega} \wedge dx = 0$). $\max|\bar{\omega}_u \cdot x_v - \bar{\omega}_v \cdot x_u|$ normalized by the max term norms. Threshold: < 0.05.

R3. Exactness. Path-dependence of $F^\pm$ integration: the integrals are computed via two routes (first u then v, and vice versa), and the max relative difference is bounded. Threshold: < 0.1.

R4. Purity ($F^\pm \in \operatorname{Im}\mathbb{H}$). $\max|\mathrm{Re}(F^\pm)| < 0.1$.

R5. Procrustes. Disparity between $F^\pm$ (retraction) and $f^\pm$ (Eq. 49), as described in section 7.4. Threshold: < 0.10.

Remark 9.2. The conformality improvement from finite-difference to analytic derivatives is a factor of 2.4×10^6, with per-seed improvements ranging from 19,700× ($k = 3$) to 1,026,000× ($k = 6$). This demonstrates the necessity of analytic derivatives for validation of the retraction form.

10 Open Problems and Conclusions

10.1 Summary of results

We have:

- extended the Bobenko–Hoffmann–Sageman-Furnas construction to $k = 3, \ldots, 1000$ (998 seeds), with step-10 sampled continuation providing corroborative evidence for the asymptotic law to $k > 6 \times 10^5$,
- obtained the mixed local-analytic/numerical asymptotic law (20) with amplitude $A = 7.081\,416\,491\,399\ldots$ (19 significant digits; (21)), a numerically observed sub-leading correction c_3/k^2 (observation 6.3), and the local amplitude formula (27),
- derived the moduli-space flow ODE (30) via implicit differentiation, reproducing the archived $k = 5, 6, 7$ sweep data to $< 0.002\%$ (section 8.1),

Table 4: Retraction form validation on 8 representative seeds (analytic derivatives, evaluated on a 120×120 grid with the legacy comparison window $\epsilon = 0.3$. All displayed entries are max-norm relative values except the Procrustes disparity. The thresholds (0.05 for R1–R2, 0.10 for R3–R5) are operational diagnostics rather than theorem-level constants. With this baseline choice, the seeds $k = 5, 7, 10, 20, 50, 100, 500$ satisfy all five gates, while $k = 3$ is borderline only in the R5 Procrustes diagnostic (0.106 against threshold 0.10). Enlarging the comparison window makes $k = 3$ pass but systematically shifts the high-k Procrustes scale, so we keep $\epsilon = 0.3$ in the main table. The Procrustes values decrease toward a small nonzero level near 0.047 as k increases.

k	R1 (closure)	R2 (cross)	R3 (exact)	R4 (purity)	R5 (Procr.)
3	3.7×10^{-2}	2.6×10^{-7}	5.6×10^{-3}	1.1×10^{-16}	0.106
5	3.6×10^{-2}	4.6×10^{-6}	5.8×10^{-3}	7.8×10^{-17}	0.077
7	3.4×10^{-2}	1.2×10^{-6}	5.7×10^{-3}	8.2×10^{-17}	0.067
10	3.2×10^{-2}	1.0×10^{-6}	5.1×10^{-3}	9.5×10^{-17}	0.059
20	2.9×10^{-2}	1.1×10^{-6}	3.9×10^{-3}	6.8×10^{-17}	0.052
50	2.7×10^{-2}	7.8×10^{-7}	2.5×10^{-3}	7.5×10^{-17}	0.048
100	2.6×10^{-2}	6.3×10^{-8}	1.8×10^{-3}	6.7×10^{-17}	0.047
500	2.7×10^{-2}	7.4×10^{-7}	7.8×10^{-4}	1.1×10^{-16}	0.046

- established the local cross-ratio asymptotic mechanism and the closure identity fixing the coefficient 4, together with finite archived root-topology certificates for the recorded B3 layer,
- formulated the conjectural spectral-modulus relation $\tau_{\text{spec}}(k) = -\frac{1}{2} + \mathrm{i}\,\ln(4k)/\pi + \mathcal{O}(k^{-2})$ (conjecture 8.22), with leading nome $q_{\text{spec}} = -\mathrm{i}/(4k)$, together with the corrected branch relation $\mathrm{CR}_k = (1 - \lambda(\tau_{\text{spec}}(k)))^{-1}$, a finite-data audit on 50 archived spectral-period seeds in the verified range $k = 5$ to 200, and a conditional explanation of $\mathrm{Re}\,\tau_{\text{spec}} = -1/2$,
- validated the retraction form on 8 representative seeds via a five-gate audit, with one low-fold case borderline only in R5 under the baseline comparison window, and benchmarked the analytic-derivative retraction pipeline on archived multi-seed validation runs,
- clarified that the theorem-level retraction object is the holomorphic distortion differential, while the implementation-side mixed-term normalization remains an audit-level statement,
- obtained PSLQ positive hits for selected secondary constants and negative searches for both A (19 digits, 37 basis families; computational certificate 6.11) and $C_2 = s_{2,\infty}$ (19 digits, 36 basis families; computational certificate 6.10): no low-height integer relation was found within the tested basis sets,
- numerical evidence that the tested fixed-τ slice of $\mathcal{M}$ follows a contractible picture: each moduli curve γ_k is a simple arc, the 998-seed cross-section is monotone and injective, and the archived inter-fold curves are nested without self-intersections (section 8.4).

10.2 Open problems

OP1. Higher-order remainder and the c_3 term. The formal derivation of section 6.5 yields $A^2 = 2\tilde{Q}_2^2\,\Delta s^2/(Z_0\,\tilde{Q}_2'\,Q_3)$, verified to 6 ppm, and derives the leading two terms of the expansion with numerically validated $\mathcal{O}(k^{-2})$ remainder (observation C.8). Extending the formal derivation to cover the numerically observed c_3/k^2 correction (observation 6.3) requires bounding the $\mathcal{O}(\delta^4)$ contributions to the period integral—a feasible but technically involved computation that we leave open. Numerical evidence (section 6.5) confirms the absence of $\mathcal{O}(\delta^2 \log \delta)$ terms: the remainder $\cdot\, k$ converges to the constant 0.10746. *Status: open.*

OP2. Nature of the amplitude A and the limit C_2. Our PSLQ analysis (computational certificates 6.10 and 6.11) and the Borodin–Okounkov bound $\mathcal{C}_{\theta,\varphi} \in (0, 1)$ (remark 6.12)

jointly exclude a direct algebraic or Fredholm-determinant encoding of A^2. Both constants carry strong computational PSLQ negatives at 19 digits: A against 37 bases (computational certificate 6.11) and C_2 against 36 bases (computational certificate 6.10), including elliptic integrals K, E, K', Eisenstein series E_2, E_4, E_6, the j-invariant, and Dedekind's η: no low-height integer relation involving A or C_2 and these constants was found. The amplitude factorises as $A^2 = (\tilde{Q}_2^2/(\tilde{Q}_2' Q_3)) \cdot 2\Delta s^2/Z_0$, separating a *spectral residue* $F_{\text{loc}} \approx 14.95$ from a *period ratio* $F_{\text{glob}} \approx 3.35$. The PSLQ negativity is consistent with the isospectral nature of the gauge flow (section 8.3): A^2 lives on a single isospectral fiber at fixed τ, not on the isomonodromic base where PVI connection coefficients reside (see OP3 below). *Status: open.*

OP3. Bridge PVI to k-fold periodicity. Dornic and Conte [7] show that PVI with Manin coefficients $[0, 0, 0, 0]$ governs generic Bonnet surfaces. Numerical computation confirms $\mathcal{C}_{\theta,\varphi} < 1$ universally, closing the direct $\ell \to k$ route via connection constants. As $k \to \infty$, $\text{CR}_k \sim e^{-4i/k} \to 1$, approaching the fixed singularity $t = 1$ of PVI. However, the k-fold gauge flow and PVI isomonodromy are structurally transversal (remarks 8.18 and 8.21): the former keeps the relevant cross-ratio numerically fixed while the latter varies it. In the logarithmic branch $\sigma_{0t} = 0$ (cf. remark 8.21), the non-trivial data are pushed into the regime (51).
Status: open, with a conjectural organizing relation (conjecture 8.22). The spectral modular parameter $\tau_{\text{spec}} = -\frac{1}{2} + \mathrm{i}\ln(4k)/\pi - \mathrm{i}/(8\pi k^2) + \mathcal{O}(k^{-4})$ is the strongest conjectural bridge presently available: it suggests, but does not yet prove, that k is encoded in the conformal type of the spectral curve via (52), and that the elliptic lambda function is numerically matched by an anharmonic transform of the branch-point cross-ratio. The real part $\text{Re}\,\tau_{\text{spec}} = -1/2$ is accounted for conditionally in conditional theorem 8.27. The factorization $A^2 = F_{\text{loc}} \times F_{\text{glob}}$ (eq. (27)) mirrors Gavrylenko–Lisovyy τ-function expansions [14, 15]. What remains open is the *connection constant* C_0 in (51): it is conjectured (conjecture 8.31) but not numerically extractable due to the logarithmic degeneracy of the unipotent PVI asymptotics.

OP4. Higher-order cross-ratio asymptotics. *Status: leading constant resolved* (propositions 8.13 and 8.15 and remark 8.16). The local branch-point calculation gives

$$\arg(\text{CR}_k)k = -4\,A^2\beta_\infty\gamma_\infty/\Delta_\infty^2 + \mathcal{O}(k^{-1}),$$

and the closure identity

$$A^2\beta_\infty\gamma_\infty/\Delta_\infty^2 = 1$$

is proved in proposition 8.15. Hence the local high-fold branch satisfies $\arg(\text{CR}_k)k = -4 + \mathcal{O}(k^{-1})$ under the chosen root ordering. What remains open is the stronger observed cancellation of the $1/k$ correction, which requires the fourth-order branch expansion.

OP5. Topology of the moduli space $\mathcal{M}$. *Status: numerical evidence only* (section 8.4). On the tested slices, the available data are numerically consistent with a contractible picture (observation 8.37): each γ_k is a simple arc terminating at a fold singularity, and the cross-section $k \mapsto (\delta_k, s_{2k})$ is a monotone simple arc with no self-intersections (observation 8.34). The same picture extends on the tested sweeps to four additional lattice parameters $\tau_{\text{im}} \in \{0.20, 0.25, 0.30, 0.35\}$ (remark 8.38), with extended δ-fold tests at k up to 100. At $\tau_{\text{im}} = 0.20$, γ_k develops δ-folds (amplitude $\sim k^{-0.22}$) but s_2-injectivity is preserved for all k tested. Bisection locates the fold onset at $\tau_{\text{fold}} = 0.24475 \pm 2 \times 10^{-6}$, establishing a bifurcation hierarchy $\tau_{\text{fold}} < \text{Im}\,\tau_{\text{ref}} < \lambda_0$. Across the archived tests, no numerical evidence against this contractibility picture is detected.

OP6. Gauge dependence of A. Continuation to 9 gauge values ($s_1 = -6, -7, \ldots, -11$) confirms gauge-dependence of A^2: it ranges from 34.3 at $s_1 = -6$ to 66.0 at $s_1 = -11$ (versus 50.1 at $s_1 = -8.5$). The closed-form formula (27) correctly predicts A^2 at all gauges. The finite-k ratio $A^2/\Delta s$, evaluated at $k = 500$, is numerically constant to 0.09% across

all 9 gauge values ($A^2/\Delta s \approx 3.1777$ at that reference k), suggesting the spectral gap $\Delta s = s_2 - s_1$ as the natural normalization. A partial algebraic explanation is the identity (remark 6.8): $\tilde{Q}_2(s_2)$ depends only on s_2 (not on s_1), providing the dominant cancellation. *Status: partially explored.*

OP7. Extension beyond genus 1. For which lattice parameters τ does the spectral curve $\mu^2 = Q(s)$ degenerate to genus 0? The degeneration locus may encode further arithmetic structure. A genus-2 extension is now known to be *obstructed* within the BHS framework: the two independent obstructions (remark 8.19) show that (i) the topological genus $g_{\text{surf}} \geq 2$ is blocked by mandatory umbilics, and (ii) the spectral genus is rigidly ≤ 1 by the Riccati/Lamé structure, even for tori. The former genus-2 test of the scaling law (35) is therefore moot: proposition 8.13 already resolves -4 as structural (pairwise root coalescence), independently of any higher-genus data. The only known compact surfaces with genus-2 spectral curves—the Wente tori [16, 10]—are CMC, and their Bonnet mates are the trivial associated-family rotations. Whether non-CMC Bonnet pairs of spectral genus > 1 can be constructed outside the class of isothermic surfaces with planar curvature lines remains an open question. *Status: partially resolved (genus-2 ruled out).*

10.3 Concluding remarks

The 998-seed dataset, together with the flow ODE, the spectral invariants, the analytic identities, and the retraction form validation, provides a computational foundation for the study of compact Bonnet pairs. The asymptotic law $\delta_k = A/\sqrt{k}$ now rests on both high-precision numerical evidence and a formal spectral-curve derivation of the amplitude. The isospectral rigidity of the cross-ratio at ω_{crit} (observation 8.10) suggests that the Bonnet moduli space carries richer integrable structure than is apparent from the periodicity conditions alone.

A Evaluated Constants

All constants are evaluated at gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$. For the theta data, the table records the real-normalized quantities used by the computational pipeline, not the raw complex Jacobi theta constants.

Symbol	Value	Description
ω	0.389018047587	Critical point: $\vartheta_2'(\omega\|\tau) = 0$
$R(\omega)$	1.374541408545	Symmetry-sphere radius (6)
U	−0.727515368968	Lamé coefficient (7)
U'	−0.403940416080	Eq. (8)
U_1'	1.262773701909	Wronskian coeff. (9)
U_2	−1.469833235911	Third Lamé coeff. (10)
$\Re\vartheta_1'(0)$	1.994641412049	Real-normalized theta-1 derivative
$\Re\vartheta_2(0)$	1.241397427680	Real-normalized theta-2 datum
$\Re\vartheta_3(0)$	1.035631327079	Real-normalized theta-3 datum
$\Re\vartheta_4(0)$	1.035631327079	Real-normalized theta-4 datum

Symbol	Value	Description
A	$7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}$	Asymptotic amplitude (Richardson)
$C_1 = A^2$	$50.146\,459\,524\,661\,3 \pm 5 \times 10^{-17}$	Squared amplitude
c_1	$0.857\,656\,047\,266\,889 \pm 5 \times 10^{-15}$	Multiplicative sub-leading coeff. k^{-1}
c_3	0.75 ± 0.01	Multiplicative sub-leading coeff. k^{-2}
$C_2 = s_{2,\infty}$	$7.323\,247\,368\,998\,298\,959 \pm 5 \times 10^{-18}$	Limiting s_2; see (25)
C_4	$2\omega - 3/10$	Moduli relation
C_5	± 4	Local cross-ratio limit
q_{spec}	$-\mathrm{i}/(4k)$	Spectral nome; see observation 8.24
C_0	$\approx 2.654\,287$	Connection constant (conjectured); see conjecture 8.31

Remark A.1. The identity $R(\omega) \cdot U(\omega) = -1$ is verified to 15 digits, consistent with (7). In the raw Jacobi convention one has $\vartheta_4(0) = \overline{\vartheta_3(0)}$ for $\Re\tau = 1/2$; the table records the common real part used in the computational normalization. For the conjectural spectral-modulus expansion $\tau_{\text{spec}} = -\frac{1}{2} + \frac{\mathrm{i}}{\pi}[\ln(4k) - \frac{1}{8k^2} + \mathcal{O}(k^{-4})]$, see conjecture 8.22 and observation 8.29.

B Reproducibility

The manuscript-specific public repository is the Compact-Bonnet-Pairs-for-k-fold-Tori repository. The archived code, manuscript source, figures, and data products cited below are intended to be read relative to that repository.

Core scripts. The main computational components cited in the body are:

- `scripts/theta_functions.py`: theta functions and spectral coefficients,
- `scripts/theorem7_periodicity.py`: Newton solver for eqs. (14) and (15),
- `scripts/retraction_form.py`: Burstall–Hoffmann–Pedit–Sageman-Furnas retraction-form construction,
- `scripts/analytic_derivatives.py`: analytic derivative pipeline,
- `scripts/frame_integrator.py`: frame ODE integration,
- `scripts/run_full_series.py`: reproduction of the 998-seed closure series,
- `scripts/generate_figures.py`: figure-PDF generation for the manuscript.

Primary archived outputs. The principal data products used in this version of the paper are:

- `results/phase15_asymptotic/full_series_k3_1000.json`: the 998 archived closure seeds,
- `results/phase15_asymptotic/critical_recheck/`: residual re-solve audit for the 48 rows of the archived 998-seed branch file with residual above 10^{-10},
- `data/high_precision_checkpoints.json`: archived 120-seed high-precision checkpoint set used for the Richardson extrapolation and the finite checkpoint validation in validated claim 6.1 and observation C.8,
- `paper/figures/*.pdf`: manuscript figures,
- `results/B3/finite_certificate/`: exact-rational finite root-topology certificates for the archived B3 seed and sweep data,
- `results/B4/high_precision/B4_full_precision_seed_data_kle200.csv` and companion summary JSON: archived full-precision B4 seed-level audit on the verified 50-seed subset in the range $k = 5$ to 200,
- `results/C2/raw_grid/`: archived raw-grid C2 normalization audits.

Reproduction commands. The exact command lines are listed in `REPRODUCE.md`. The core manuscript artifacts are reproduced by the scripts:

- `scripts/run_full_series.py` writes the main seed series,
- `scripts/export_phase15_critical_recheck.py` regenerates the critical-row residual audit for the archived 998-seed branch,
- `scripts/generate_figures.py` writes the figure PDFs,
- `scripts/export_B3_certificate_inputs.py` regenerates the archived B3 certificate inputs,
- `scripts/export_B4_high_precision_seed_data.py` regenerates the archived B4 high-precision seed dataset used for the finite-data audit, and
- `scripts/export_C2_raw_grid_audit.py` regenerates the archived C2 raw-grid audits.

Environment. The current project reproducibility guide `REPRODUCE.md` lists Python 3.10+ with `numpy`, `scipy`, and `mpmath` as the core runtime requirements; figure generation additionally needs `matplotlib`. The archived high-precision computations use 40-digit or higher `mpmath` arithmetic with fixed-order Gauss–Legendre quadrature. The exact repository revision used for a public release should be recorded in the accompanying release tag or archive manifest.

C Formal asymptotics for $\delta = A/\sqrt{k}$

This appendix records the three estimates announced in remark 6.9 in the form used in the paper. Together with the archived numerical checks, they support the formal expansion of section 6.5 and its stated remainder behaviour on the tested branch. All numerical constants below refer to the gauge $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$.

Setup. Recall that $\tilde{Q}_2(s)$ is a *quadratic* polynomial in the integration variable s (at fixed parameters s_1, s_2). Writing

$$\tilde{Q}_2(s) \;=\; P\,s^2 - 2M\,s - 1, \tag{65}$$

where

$$P = U^{-2}\big(2(s_1+s_2)\,U_1' - U_2\big), \tag{66}$$

$$M = U^{-2}\big(U' + s_1\,s_2\,U_1'\big), \tag{67}$$

we have $Z_0^2 = P + M^2$ (at $\delta = 0$), and the cancellation of s_1-dependent terms at $s = s_2$ (remark 6.8) follows from $P\,s_2^2 - 2M\,s_2 = U^{-2}(2U_1's_2^3 - U_2s_2^2 - 2U's_2)$.

At $\delta = 0$ with $s_2 = s_{2,\infty}$: $\tilde{Q}_2(s_{2,\infty}) = 2033.2$, $\tilde{Q}_2' = 2Ps_2 - 2M = 257.0$, $\tilde{Q}_2'' = 2P = -5.676$.

Lemma C.1 (Bracket linearization)**.** *For $|s - s_{2,\infty}| \le a$ with $a \le a_{\max} := \delta_{100}\sqrt{Q_3(s_{2,\infty})}/|\Delta s| \approx 1.457$, the reciprocal $f(s) = 1/\tilde{Q}_2(s)$ satisfies*

$$\Big|f(s) - f(s_{2,\infty}) - f'(s_{2,\infty})(s - s_{2,\infty})\Big| \;\le\; \frac{L_a}{2}\,(s - s_{2,\infty})^2,$$

where $L_a = \max_{|t-s_2|\le a_{\max}} |f''(t)| = 3.32 \times 10^{-5}$.

Proof. Since $\tilde{Q}_2$ is quadratic, $f = 1/\tilde{Q}_2$ is C^∞ in any region where $\tilde{Q}_2 > 0$. On the interval $[s_{2,\infty} - a_{\max},\, s_{2,\infty} + a_{\max}]$ we have $\min \tilde{Q}_2 \ge 1652.8 > 0$, so f is smooth and Taylor's theorem with Lagrange remainder gives the bound. The second derivative is $f'' = (2(\tilde{Q}_2')^2 - \tilde{Q}_2\,\tilde{Q}_2'')/\tilde{Q}_2^3$ with $\tilde{Q}_2'' = 2P$ constant; evaluation on a 201-point grid gives $L_a = 3.32 \times 10^{-5}$.

Parity. Write $\bar{s} = (s_2^+ + s_2^-)/2$ for the oval midpoint (cf. lemma C.3). The constant shift $f(\bar{s}) - f(s_2)$ multiplies $\int Q_2\,ds/\sqrt{Q} = F_{\mathrm{ax}} = 0$ and drops out, so we may center the bracket at $\bar{s}$. Under $s = \bar{s} + a\sin\varphi$, the remainder $f(s) - f(\bar{s}) - f'(\bar{s})(s - \bar{s}) = \mathcal{O}(a^2\sin^2\varphi)$ is even in φ, while

the odd part of Q_2 integrates against this even remainder to give zero. The surviving (odd×odd) terms are $\mathcal{O}(\delta^4)$, so the bracket linearization contributes $\mathcal{O}(\delta^4)$ to $\theta/2$, not $\mathcal{O}(\delta^3)$. Explicitly:

$$|\Delta I_a| \;\le\; C_a\,\delta^4, \qquad C_a = \frac{Z_0\,L_a\,Q_3(s_{2,\infty})^2\,3\pi}{16\,\Delta s^4} = 0.054.$$

□

Lemma C.2 (Pinching-radius asymptotics)**.** *Under the hypotheses of lemma 8.12, the real oval of $Q(s)$ near s_2 has endpoints*

$$s_2^-(\delta) = m_2(\delta) - h_2(\delta), \qquad s_2^+(\delta) = m_2(\delta) + h_2(\delta).$$

Hence its midpoint $\bar{s} = (s_2^+ + s_2^-)/2$ and half-width $a = (s_2^+ - s_2^-)/2$ satisfy

$$\bar{s} = s_2 + \left(\frac{Q_3'(s_2)}{2(\Delta s)^2} - \frac{Q_3(s_2)}{(\Delta s)^3}\right)\delta^2 + \mathcal{O}(\delta^4), \tag{68}$$

$$a = \frac{\delta\sqrt{Q_3(s_2)}}{\Delta s} + \mathcal{O}(\delta^3). \tag{69}$$

Equivalently, if $a_0 = \delta\sqrt{Q_3(s_2)}/\Delta s$, then

$$a = a_0(1 + \mathcal{O}(\delta^2)).$$

The one-sided distances are

$$s_2^+ - s_2 = a_0 + \alpha_2\delta^2 + \mathcal{O}(\delta^3), \qquad s_2 - s_2^- = a_0 - \alpha_2\delta^2 + \mathcal{O}(\delta^3),$$

with

$$\frac{\alpha_2}{\beta} = \frac{\beta}{2}\left(\frac{Q_3'(s_2)}{Q_3(s_2)} - \frac{2}{\Delta s}\right), \qquad \beta = \frac{\sqrt{Q_3(s_2)}}{\Delta s}.$$

The estimates are uniform on compact parameter sets satisfying the sign and separation hypotheses of lemma 8.12.

Proof. This is exactly the real part of lemma 8.12. The endpoints of the real oval are the two real roots near s_2, namely $m_2 \mp h_2$. Therefore the midpoint is m_2 and the half-width is h_2. Substituting the expansions

$$m_2 = s_2 + \alpha_2\delta^2 + \mathcal{O}(\delta^4), \qquad h_2 = \beta\delta + \mathcal{O}(\delta^3)$$

gives (68)–(69). The one-sided formulas follow by adding and subtracting the midpoint displacement. Finally,

$$\frac{\alpha_2}{\beta} = \frac{1}{\beta}\left(\frac{Q_3'(s_2)}{2(\Delta s)^2} - \frac{Q_3(s_2)}{(\Delta s)^3}\right) = \frac{\beta}{2}\left(\frac{Q_3'(s_2)}{Q_3(s_2)} - \frac{2}{\Delta s}\right).$$

□

Lemma C.3 (Endpoint parity and desingularization)**.** *Let δ be a signed local parameter and put $\eta = \delta^2$. Near the real oval around s_2, the two real branch points have the parity form*

$$s_2^{\pm}(\delta, s) = \bar{s}(\eta, s) \pm \delta\,\widehat{a}(\eta, s), \qquad \widehat{a}(0, s) = \frac{\sqrt{Q_3(s)}}{s - s_1},$$

after choosing the sign of δ so that $s_2^+ > s_2^-$ for $\delta > 0$. Consequently the positive half-width is not analytic in η by itself; rather

$$a(\delta, s) = \frac{s_2^+(\delta, s) - s_2^-(\delta, s)}{2} = \delta\,\widehat{a}(\delta^2, s) \qquad (\delta > 0).$$

Under the substitution

$$s = \bar{s}(\eta, s_2) + \delta\,\widehat{a}(\eta, s_2)\sin\varphi, \qquad -\frac{\pi}{2} \le \varphi \le \frac{\pi}{2},$$

the square-root endpoint singularities are removed algebraically:

$$\frac{\mathrm{d}s}{\sqrt{Q(s)}} = \frac{\mathrm{d}\varphi}{\sqrt{R(s,\delta)}}, \qquad R(s,\delta) = \frac{-Q(s,\delta)}{(s - s_2^-(\delta, s_2))(s - s_2^+(\delta, s_2))}. \tag{70}$$

The pulled-back integrands entering the local period expansion are real analytic in (φ, δ, s_2) for small δ and have no endpoint logarithms. Pointwise analyticity in η is not asserted at this stage.

Proof. The local branch-point splitting from lemma 8.12, applied with s_2 near the pinching centre, gives

$$s_2^{\pm}(\delta, s_2) = s_2 + \alpha_2(s_2)\delta^2 \pm \beta(s_2)\delta + \mathcal{O}(\delta^3), \qquad \beta(s_2) = \frac{\sqrt{Q_3(s_2)}}{s_2 - s_1}.$$

Analytic perturbation after the square-root blow-up at the double root gives the refined parity representation $s_2^{\pm} = \bar{s}(\eta, s_2) \pm \delta\widehat{a}(\eta, s_2)$.

With the signed parametrization above,

$$(s - s_2^-)(s - s_2^+) = -\delta^2\widehat{a}(\eta, s_2)^2\cos^2\varphi,$$

so the factor $\cos\varphi$ in $\mathrm{d}s$ cancels the square-root singularity at both endpoints. The remaining denominator $R(s,\delta)$ is non-zero and positive on the compact integration interval for sufficiently small δ, because the other two roots stay separated near s_1. The remaining theta–rational factors are rational functions of the analytic quantity $\bar{s} + \delta\widehat{a}\sin\varphi$ and have no poles on the oval for the reference branch. Thus the pulled-back integrands are analytic on a fixed compact φ-interval. This algebraic desingularization rules out $\delta^m \log\delta$ endpoint terms in the local period expansions. □

Lemma C.4 (Evenness of the reduced period maps)**.** *Let $u = \sin\varphi$ and suppose a desingularized integrand $H(u, \delta, s)$ is analytic for $u \in [-1, 1]$ and small δ, and satisfies the involutive symmetry*

$$H(-u, -\delta, s) = H(u, \delta, s). \tag{71}$$

Then

$$I(\delta, s) = \int_{-\pi/2}^{\pi/2} H(\sin\varphi, \delta, s)\,\mathrm{d}\varphi$$

is an even analytic function of δ, hence can be written as a real-analytic function of $\eta = \delta^2$.

For the Bonnet period maps used below, the axial reduced period $\mathcal{X}$ and the axial-subtracted half-angle Θ satisfy this symmetry after the removable factor η is divided out in F_{ax} and after the constant axial term is subtracted in the half-angle. Consequently

$$\mathcal{X}(\eta, s) = X_0(s) + \eta X_1(s) + \eta^2 X_2(s) + \cdots,$$
$$\Theta(\eta, s) = \eta\Theta_1(s) + \eta^2\Theta_2(s) + \eta^3\Theta_3(s) + \cdots$$

are real analytic in (η, s) near $(0, C_2)$.

Proof. By analyticity, write locally

$$H(u, \delta, s) = \sum_{m,n\ge 0} h_{mn}(s)\,u^m\delta^n.$$

Condition (71) forces $h_{mn} = 0$ unless $m + n$ is even. Hence every term with odd n has odd m, and its contribution integrates to zero over the symmetric interval because $\sin^m \varphi$ is odd. Therefore only even powers of δ remain in $I(\delta, s)$.

In the present Bonnet integrands, changing δ to $-\delta$ swaps the two real branch points and is compensated by $u \mapsto -u$ in the parametrization $s = \bar{s} + \delta\widehat{a}u$. The quartic Q and the spectral coefficients depend on δ through δ^2, while the only signed dependence arises through the factor $\delta\widehat{a}u$. Thus odd powers of δ occur paired with odd powers of u. The axial subtraction in the half-angle removes the constant axial contribution before division by the leading small parameter, so the same symmetry applies to the reduced half-angle. The previous lemma supplies analyticity and absence of logarithmic endpoint terms; the parity argument then gives analytic descent from δ to $\eta = \delta^2$. □

Definition C.5 (Reduced period map near the pinched oval)**.** Set

$$\eta = \delta^2, \qquad \varepsilon = \frac{1}{k}.$$

Let C_2 be a candidate real pinching centre and let s vary in a small neighbourhood of C_2. For $\delta > 0$ sufficiently small, let $s^-(\delta, s) < s^+(\delta, s)$ be the two real branch points near s. By lemma C.3 these can be written in the parity form

$$s^{\pm}(\delta, s) = \bar{s}(\eta, s) \pm \delta\, \widehat{a}(\eta, s), \qquad \widehat{a}(0, s) = \frac{\sqrt{Q_3(s)}}{s - s_1}.$$

Thus the half-width is

$$a(\delta, s) = \delta\, \widehat{a}(\eta, s);$$

it is not analytic in η before this factor of δ is separated.

After the substitution

$$s' = \bar{s}(\eta, s) + \delta\, \widehat{a}(\eta, s) \sin\varphi,$$

the square-root endpoint singularities in both period integrals are removed as in (70). The pulled-back integrands are analytic in (φ, δ, s) by lemma C.3; the parity descent in lemma C.4 is what makes the reduced integrated period maps below real analytic in (η, s).

We denote by

$$\mathcal{X}(\eta, s)$$

the desingularized axial period with its removable factor of η divided out, so that

$$F_{\mathrm{ax}}(\sqrt{\eta}, s) = \eta\, \mathcal{X}(\eta, s),$$

and by

$$\Theta(\eta, s)$$

the axial-subtracted half-angle appearing in section 6.5. Thus, on the axial locus, the closing condition is

$$\Theta(\eta, s) = \frac{\pi}{k} = \pi\varepsilon.$$

Their expansions have the form

$$\mathcal{X}(\eta, s) = X_0(s) + \eta X_1(s) + \eta^2 X_2(s) + \cdots, \tag{72}$$

$$\Theta(\eta, s) = \eta\Theta_1(s) + \eta^2\Theta_2(s) + \eta^3\Theta_3(s) + \cdots. \tag{73}$$

Moreover, at the reference pinching centre,

$$\Theta_1(C_2) = \frac{Z_0\, \tilde{Q}_2'(C_2)\, Q_3(C_2)\, \pi}{2\, \tilde{Q}_2(C_2)^2\, (C_2 - s_1)^2}. \tag{74}$$

Theorem C.6 (Local high-fold branch near the pinched oval)**.** *Let C_2 be a real zero of the reduced axial leading term satisfying*

$$X_0(C_2) = 0, \qquad X_0'(C_2) \neq 0, \tag{75}$$

and assume also

$$\Theta_1(C_2) > 0. \tag{76}$$

Then there are $\varepsilon_0 > 0$ and analytic functions

$$\eta(\varepsilon), \qquad s(\varepsilon), \qquad 0 \leq \varepsilon < \varepsilon_0,$$

with $\eta(0) = 0$ and $s(0) = C_2$, such that

$$F_{\mathrm{ax}}(\sqrt{\eta(\varepsilon)}, s(\varepsilon)) = 0, \qquad \Theta(\eta(\varepsilon), s(\varepsilon)) = \pi\varepsilon. \tag{77}$$

Consequently, for every integer $k > 1/\varepsilon_0$ there is a local high-fold solution

$$\delta_k = \sqrt{\eta(1/k)}, \qquad s_{2,k} = s(1/k),$$

and it satisfies

$$\delta_k = \frac{A}{\sqrt{k}}\left(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\right), \tag{78}$$

$$s_{2,k} = C_2 + \frac{\sigma_1}{k} + \mathcal{O}(k^{-2}). \tag{79}$$

The leading amplitude is determined by the desingularized angle coefficient:

$$A^2 = \frac{\pi}{\Theta_1(C_2)} = \frac{2\,\tilde{Q}_2(C_2)^2\,(C_2 - s_1)^2}{Z_0\,\tilde{Q}_2'(C_2)\,Q_3(C_2)}. \tag{80}$$

More explicitly, if

$$s(\eta) = C_2 + \rho_1\eta + \mathcal{O}(\eta^2), \qquad \rho_1 = -\frac{X_1(C_2)}{X_0'(C_2)},$$

and

$$D_* := \Theta_2(C_2) + \Theta_1'(C_2)\rho_1, \tag{81}$$

then

$$c_1 = \frac{D_* A^4}{2\pi}, \qquad \sigma_1 = \rho_1 A^2. \tag{82}$$

Proof. By definition C.5, the axial condition near the pinched oval is equivalent to

$$\mathcal{X}(\eta, s) = 0.$$

The first non-degeneracy condition in (75) therefore allows the implicit function theorem to be applied at $(\eta, s) = (0, C_2)$. Hence there is a unique analytic axial branch

$$s = s(\eta) = C_2 + \rho_1\eta + \mathcal{O}(\eta^2), \qquad \rho_1 = -\frac{X_1(C_2)}{X_0'(C_2)}.$$

Substituting this branch into the reduced half-angle gives

$$\Theta(\eta, s(\eta)) = \Theta_1(C_2)\eta + (\Theta_2(C_2) + \Theta_1'(C_2)\rho_1)\eta^2 + \mathcal{O}(\eta^3).$$

Set $B := \Theta_1(C_2)$ and $D_* := \Theta_2(C_2) + \Theta_1'(C_2)\rho_1$. Since $B > 0$, the equation

$$B\eta + D_*\eta^2 + \mathcal{O}(\eta^3) = \pi\varepsilon$$

has, again by the implicit function theorem, a unique analytic solution

$$\eta(\varepsilon) = \frac{\pi}{B}\varepsilon - \frac{D_*\pi^2}{B^3}\varepsilon^2 + \mathcal{O}(\varepsilon^3).$$

Writing $A^2 = \pi/B$ gives

$$\eta(\varepsilon) = A^2\varepsilon - \frac{D_*A^6}{\pi}\varepsilon^2 + \mathcal{O}(\varepsilon^3).$$

Taking the positive square root yields

$$\delta(\varepsilon) = A\varepsilon^{1/2}\left(1 - \frac{D_*A^4}{2\pi}\varepsilon + \mathcal{O}(\varepsilon^2)\right),$$

which is (78) with $c_1 = D_*A^4/(2\pi)$ after setting $\varepsilon = 1/k$. The expansion for $s_{2,k}$ follows from $s(\eta(\varepsilon)) = C_2 + \rho_1 A^2\varepsilon + \mathcal{O}(\varepsilon^2)$. Finally, substituting the explicit value (74) of $\Theta_1(C_2)$ gives the amplitude formula (80). □

Remark C.7 (What is proved locally and what remains computational). Theorem C.6 is a local analytic theorem: once a simple zero C_2 of the reduced axial leading term is fixed and the leading angle coefficient is positive, the high-k branch and the $k^{-1/2}$ law follow from the implicit function theorem. The theorem then implies $s(\varepsilon) \to C_2$ as $\varepsilon \to 0$; it should not be read as defining C_2 circularly from a pre-existing numerical branch.

The theorem does not by itself identify this local branch with the entire numerical continuation branch for $k = 3, \ldots, 1000$; that requires a global no-branch-jump argument or a certified continuation audit. In the present manuscript, the non-degeneracy conditions and the finite-branch identification remain supported by the numerical validations described in sections 5.3, 6.1 and 6.4.

Observation C.8 (Checkpoint validation of the numerical remainder)**.** For the computed reference branch at $s_1 = -8.5$, $\tau = \frac{1}{2} + 0.3205128205\,\mathrm{i}$, the local expansion of theorem C.6 is numerically realized on the checkpoint data with

$$\delta_k = \frac{A}{\sqrt{k}}\left(1 - \frac{c_1}{k} + \mathcal{O}(k^{-2})\right), \tag{83}$$

where

$$A = 7.081\,416\,491\,399\,252\,576 \pm 3 \times 10^{-18}, \qquad c_1 = 0.857\,656\,047\,266\,889.$$

Equivalently, on the tested checkpoint set,

$$\left|\delta_k^2 k - A^2\left(1 - \frac{2c_1}{k}\right)\right| \le \frac{C}{k}$$

with the conservative empirical constant $C \le 6$.

Proof. The analytic mechanism is given by theorem C.6; the constants are then evaluated on the computed reference branch. The explicit residual bound $C \le 6$ is not derived analytically. It is obtained by numerical validation on the checkpoint set. The ratio

$$\left|\delta_k^2 k - A^2\left(1 - \frac{2c_1}{k}\right)\right| k/C$$

stays below the reported audit threshold on the stored checkpoint data, giving an effective safety factor. The full-rank Jacobian and the absence of a branch jump are likewise verified on the tested branch rather than proved uniformly. To promote this observation to a proof-level finite certificate, the checkpoint table, script version, output log, and hashes of the input data would need to be archived with the manuscript. □

Remark C.9. The constant $C = 6$ is conservative; numerically $C \approx 1.12$ suffices. A fully analytic value of C would require the next coefficient D_* in (81), together with uniform bounds for the third-order desingularized period coefficients. This is finite but algebraically longer than the local existence theorem above.

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